

Chinese Twin Agricultural Tariffs Impact on Dairy Supply

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Abstract

This paper develops a “Twin Tariffs” theory, in which China imposes tariffs on feed inputs and dairy products. The retaliatory tariffs on an upstream feed input (alfalfa hay) and a downstream output (dairy products) jointly affect import adjustments and the costs of domestic dairy products in China. Tariff increases are associated with lower Chinese imports from the United States, with stronger adjustments in the alfalfa-related channel than in the dairy-output channel. We use a time-series growth response model with fixed effects to estimate dynamic elasticities and lagged pass-through from tariff changes to import quantities and domestic farm-gate milk prices, and a large-country partial-equilibrium twin tariff framework to interpret terms-of-trade and welfare implications under alternative tariff and subsidy scenarios. Chinese domestic milk prices increased by 0.52% relative to the baseline period (2017, prior to the trade war) during the initial trade war period (2018-2022), driven by higher costs of imported alfalfa and alternative dairy products. Welfare simulations indicate that the joint tariff regime generates larger losses than either tariff alone, while domestic support targeted at the upstream feed constraint can substantially reduce those losses.

Keywords: Tariffs; Alfalfa; Dairy; Trade War; Vertical Supply Chain

JEL Codes: F13, Q17, Q18

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1 Introduction

Trade barriers distort resource allocation, impacting producer viability, consumer welfare, and broader food security (Interesse, 2024; Fajgelbaum et al., 2024; Staples and McCullough, 2025). Yet, the majority of the agricultural trade literature evaluates these disruptions exclusively through the lens of single-commodity incidence: a tariff raises domestic prices, reduces imports, and generates deadweight loss (DWL). What remains largely unexplored is how trade barriers compound when applied across vertically linked supply chains. The 2018 U.S.-China trade war provides a prime, unexamined case study of a policy shock that we term "twin tariffs" —the simultaneous imposition of duties on both upstream inputs and downstream products. Specifically, China enacted retaliatory tariffs exceeding 25% on U.S. agricultural goods, affecting 86.3% of HS 6-digit products in 2018 (Fajgelbaum and Khandelwal, 2022; Qian, 2023; Xu et al., 2025). In the dairy sector, these tariffs targeted U.S. alfalfa hay—a critical feed input—alongside processed dairy products—the final output. This dual-targeting created cost pressures and cross-market distortions that conventional, single-commodity analyses fundamentally fail to capture.

Motivated by the failure of single-commodity models to capture these compounding distortions and structural asymmetry in the twin-tariff case, this paper investigates the mechanisms and consequences of twin-tariff regimes through four key inquiries. We first examine how simultaneous tariffs on an upstream input and a downstream output propagate through a vertically linked agricultural sector. Second, we analyze the role of asymmetric trade elasticities and substitution possibilities in determining whether the trade-war burden is absorbed through price-margin adjustments (alfalfa) or quantity-margin shifts (dairy). Third, we quantify the net welfare consequences of this joint regime, testing whether cross-market distortions lead to "super-additive" losses that exceed the sum of isolated tariff effects. Finally, we evaluate the efficacy of targeted domestic policies—such as upstream production subsidies—in offsetting these distortions.

Besides, rising incomes and urbanization have increased Chinese dairy consumption and

demand (Appendix Figure A2). And the expansion of a commercial dairy sector depends on reliable access to high-quality, protein-dense feed inputs, particularly alfalfa and soybean meal. Given China’s limited arable land relative to its population, disruptions to feed imports can directly affect the cost and stability of domestic dairy production. Tariff and subsidy policies targeting dairy products and alfalfa therefore carry broader implications for protein nutrition, agricultural resilience, and food security (Nirmal et al., 2025; Long et al., 2025).

To answer these questions, we develop a formal two-market partial equilibrium model with an explicit vertical cost linkage. We derive propositions regarding cross-market tariff pass-through, asymmetric import substitution, and the welfare amplification unique to twin tariffs. Utilizing monthly bilateral trade and price series from 2005 to 2025, we empirically estimate four model specifications that examine the linkage between tariff shocks, trade flows, and domestic prices, with a focus on the "additionality" of twin-tariff effects.

Our results yield four primary findings corresponding to our research questions. First, regarding propagation, we find that alfalfa tariffs are positively associated with increases in China’s farm-gate milk price, confirming a direct cost pass-through from imported feed to domestic production. By contrast, the dairy tariff does not generate a comparably robust domestic price effect, supporting our argument that upstream input tariffs can exert more pressure on domestic producer prices than downstream output tariffs in vertically linked systems. Second, we identify that the burden of the trade war is divided asymmetrically based on trade elasticities. For dairy products, the shock was absorbed primarily on the quantity margin; cumulative imports fell sharply as buyers successfully substituted toward alternative origins like New Zealand and the EU. For alfalfa, however, the lack of viable global substitutes meant the shock was absorbed on the price margin. This "scissors effect"—where input costs rise while output prices remain constrained by import competition—directly drives the compression of domestic producer margins. Third, we find that the joint twin-tariff regime generates a net welfare loss of \$11.7 billion, which exceeds the sum of the alfalfa-only (\$0.4B) and dairy-only (\$8.7B) counterfactuals by approximately \$2.5 billion. Our structural

model reveals that this "excess loss" is entirely driven by the cross-market distortion: the upstream alfalfa tariff shifts the downstream dairy supply curve inward, amplifying the deadweight loss generated by the direct dairy tariff. Lastly, we find that the negative effects of twin tariffs are highly sensitive to the domestic policy environment and evolve over time. While the first trade war period (2018–2022) saw a 0.52% rise in milk prices, the effect attenuated in the second period (2023–2025) as China improved its "outside options" through origin diversification. Furthermore, we find that a counterfactual domestic alfalfa producer subsidy—targeting the inelastic upstream input—reduces the net welfare loss by 76% (from \$11.7B to \$2.8B), proving far more efficient than equivalent protection of the downstream output.

Our analysis contributes to the literature in three ways. First, we develop the first partial-equilibrium model of simultaneous upstream-input and downstream-output tariffs in a vertically linked agricultural supply chain, deriving explicit propositions for cross-market pass-through and welfare "super-additivity." While existing research on the U.S.–China trade war documents own-tariff pass-through to import prices ([Amiti et al., 2019](#)) and bilateral trade reductions ([Fajgelbaum et al., 2024](#)), our framework is the first in the agricultural trade literature to formalize the joint upstream–downstream interaction or quantify the resulting cross-market welfare amplification in a large-country setting. Second, we provide the first empirical decomposition of feed-to-dairy tariff transmission that distinguishes pure origin reallocation from residual price-premium deadweight loss. While prior empirical studies analyze the alfalfa ([Hou et al., 2024](#)) and dairy ([Zhang and Abler, 2023](#)) channels in isolation, and the broader tariff-incidence literature ([Fajgelbaum and Khandelwal, 2022](#); [Chor and Li, 2024](#)) focuses on the 2018–2019 window, no prior work has jointly decomposed both channels across the 2018–2022 and 2023–2025 episodes. Our 2005–2025 span identifies a dynamic adaptation between trade war phases—a cross-window pattern that single-episode designs cannot recover. Third, we provide actionable guidance for policy design, establishing that targeted upstream intervention is substantially more efficient than equivalent downstream

protection. While existing work treats subsidies as standalone fiscal instruments ([Farrokhi and Pellegrina, 2023](#); [Schmitz et al., 2022](#); [Gagné and Gouel, 2022](#)), we quantify their welfare-offset value as a specific counterfactual to a tariff regime. Our findings establish that policy effectiveness in vertically integrated settings must be evaluated against an evolving elasticity environment rather than a static baseline.

2 Background and Conceptual Framework

2.1 Dairy and alfalfa trade

China’s dairy sector has experienced steady per capita consumption growth, driven by urbanization, rising incomes, and government nutrition campaigns emphasizing the role of protein and calcium intake ([Gooch et al., 2017](#))¹. While per capita consumption remains on a steady upward trajectory, domestic production has struggled to keep pace. Output growth is significantly hindered by land constraints and escalating feed costs, particularly within the "Golden Milk Source Belt" of Inner Mongolia, where accounts for about 20% of China’s domestic milk production (Appendix Figure [A3](#)).

The gap between domestic supply and demand has made China the second-largest milk importer worldwide. In 2024, China imported \$1.42 billion worth of milk and became the second largest importer of milk in the world. The United States serves as the third largest supplier of dairy products to China, primarily providing whey, milk powder, and lactose—all primary processed dairy lines (Figure [A3](#)) ([Grant et al., 2025](#)). While New Zealand and the European Union remain the primary sources of China’s liquid milk imports, the U.S. dominates the feed market: it accounted for 89.9% of China’s alfalfa hay imports in 2023([Hou et al., 2024](#)).

Alfalfa is the third most valuable cash crop in the U.S., with an estimated economic value of \$8.7 billion in 2023 ([U.S. Department of Agriculture, Economic Research Service \(USDA](#)

¹We illustrate the long-run trajectory of China’s per capita dairy consumption in Appendix Figure [A2](#)

ERS), 2024). It is also recognized as one of the most important forage globally due to its high nutritional value for livestock, making it a vital export commodity (Cantor et al., 2022).

2.2 The "twin tariff" intervention

The 2018 U.S.-China trade conflict introduced a distinct policy phenomenon defined here as "twin tariffs": the simultaneous imposition of retaliatory tariffs on upstream production inputs (forage) and downstream final products (dairy) (USDA FAS, 2023a). This dual-layered intervention has evolved through several distinct phases, creating a unique environment for analyzing supply chain shocks.

Phase I: Initial Escalation (2018–2019): The conflict began in July 2018, when China imposed an initial 25% retaliatory tariff on U.S. agricultural products, including alfalfa and a broad range of dairy products (Xu et al., 2025; USDA ERS, 2024). By 2019, the effective tariff burden intensified; cheese products faced total duties of 33–37%, representing a combination of Most-Favored-Nation (MFN) rates and Section 301 retaliation Bown (2021); USDA ERS (2024). Subsequent rounds in September 2019 pushed rates for fresh and grated cheese to 38–42%, while fermented milk and buttermilk reached levels as high as 50%.

Phase II: Managed Retaliation and Exclusions (2020–Early 2025): Following the Phase One Agreement in February 2020, the policy landscape shifted from blanket escalation to a regime of "managed retaliation." While retaliatory tariffs were not fully rescinded, China introduced a robust exclusion process that bifurcated the impact on the supply chain: 1) upstream product: alfalfa was granted exclusion status as early as September 2019, while whey intended for feed use received repeated extensions through February 28, 2025. 2) downstream products: Section 301 tariffs remained active for most final dairy products despite expanded market access for U.S. facilities (U.S. Department of Agriculture, Foreign Agricultural Service (USDA FAS), 2020; Xu et al., 2025). Exclusions in this category were granted only sparingly on a case-by-case basis, primarily to alleviate acute cost pressures within the domestic dairy-feed system.

Phase III: The 2025 Escalation and Geneva Negotiations: The trade environment experienced renewed volatility in early 2025, with tariffs escalating significantly. However, following the Geneva negotiations in May 2025, China implemented a temporary 90-day reduction of the additional retaliatory duty—dropping it from 125% to 10% [Wen \(2025\)](#). Despite this recent reduction, earlier tariff layers and product-specific measures persisted, resulting in a heterogeneous effective tariff burden across different agricultural categories.

2.3 China’s diversification and mitigation policy

To mitigate cost shocks and stabilize farm-gate milk prices, China introduced producer subsidies for local forage growers and expanded support for regional dairy production. Beyond domestic subsidies, China has actively pursued trade diversification to reduce its reliance on U.S. China has systematically sought alternative markets for its agricultural needs ([Qian, 2023](#); [Rose, 2024](#)). This strategy is further evidenced by China’s ratification of the Regional Comprehensive Economic Partnership (RCEP)—the world’s largest free trade area—and its application to the Comprehensive and Progressive Trans-Pacific Partnership (CPTPP) ([Rose, 2024](#); [Li et al., 2020](#)).

These diversification efforts are designed to diminish the impact of future U.S. tariffs and weaken American trade leverage. For example, China’s imports of alfalfa meal and pellets totaled 15.12 million kg in 2023, primarily sourced from Italy and Spain; imports from the U.S. fell to just 24,745 kg, a decline of approximately 90% compared to 2022. Additionally, alfalfa hay import prices dropped from \$596/ton to \$400/ton between 2022 and 2023 ([USDA ERS, 2024](#); [USDA FAS, 2023b](#)). During this same period, domestic milk production grew for four consecutive years, reaching 41.97 million tons in 2023 ([Morgan et al., 2022](#); [Steinberg and Tan, 2024](#)).

Despite these shifts, a significant dependency remains: the United States continues to be China’s principal supplier of alfalfa hay, providing 898,186 tons and accounting for 89% of that specific market in 2023 ([Qian, 2023](#)) (Appendix Figure A4).

2.4 Schematic of the twin-tariff regime

The 2018–2025 twin-tariff episode produced three observable patterns that motivate the formal analysis in the theoretical model section. First, when Chinese tariffs raised the landed price of U.S. alfalfa, imports contracted on the U.S. margin and only partially recovered through alternative suppliers, leaving Chinese dairy farms facing higher effective feed costs. Second, when tariffs were extended to U.S. dairy products, import volumes fell rapidly but were largely replaced by New Zealand and EU sourcing, so the dairy-side price pressure on Chinese consumers was muted. Third, the two adjustments moved in opposite directions for the same policy regime: the upstream shock bit on prices while the downstream shock bit on quantities.

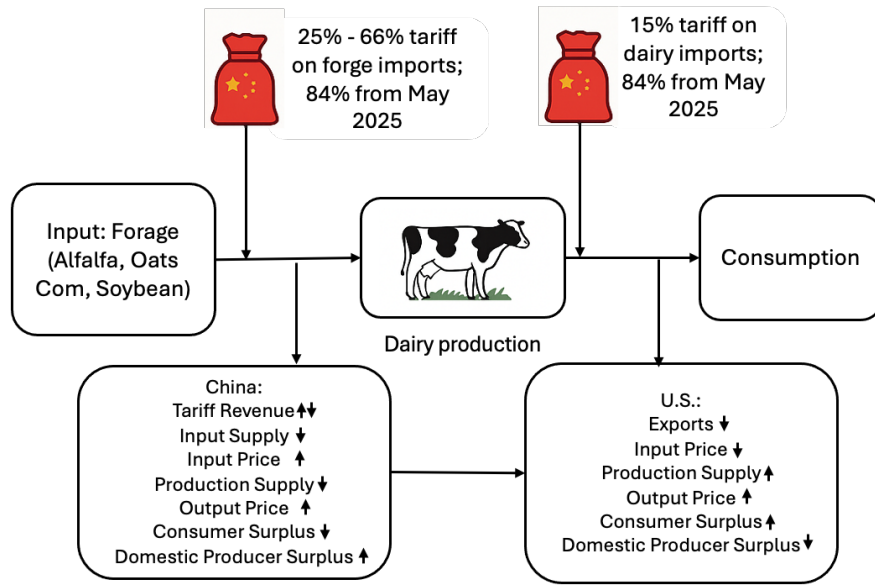


Figure 1: Framework of Chinese tariff on U.S. alfalfa and dairy impacts

Figure 1 represents the supply chain as a horizontal flow across upstream input production, midstream dairy production, and downstream consumption, with arrows indicating the asymmetric spillovers across borders. In the United States, the loss of foreign demand triggers a forage surplus, lowering input prices and the operating costs of domestic dairy producers; expanded dairy output then transfers surplus from producers to consumers (Villoria, 2026). In China, the tariffs restrict upstream supply, raise input prices, and force a contraction

in domestic dairy production—compounded by ambiguous tariff revenue from a shrinking import base (Santeramo et al., 2024). The combined shocks reduce Chinese consumer surplus while raising producer surplus, with net welfare effects that depend on the elasticities derived.

3 Theoretical Model

We develop a twin-tariff model that imposed tariffs simultaneously on an upstream input (alfalfa) and a downstream output (dairy).

3.1 Model setup

Consider two vertically linked markets: alfalfa (the upstream input, denoted by subscript a) and dairy (the downstream output, denoted by subscript d). China is modeled as a large importer in both markets. Let P_j represents the domestic price in China, P_j^* the world (export) price, and τ_j the ad valorem tariff rate on good $j \in \{a, d\}$. The domestic price is determined by the tariff-inclusive world price:

$$P_j = (\tau_j + 1) P_j^*. \quad (1)$$

China’s import demand for alfalfa is $M_a(P_a)$, where $M'_a < 0$. The rest-of-world (ROW) export supply is $X_a(P_a^*)$, with $X'_a > 0$. Under the large-country assumption, the export supply curve is upward-sloping rather than perfectly elastic.

China’s import demand for dairy products depends on both the dairy price and the price of the primary forage input, alfalfa:

$$M_d = M_d(P_d, P_a), \quad \frac{\partial M_d}{\partial P_d} < 0, \quad \frac{\partial M_d}{\partial P_a} > 0. \quad (2)$$

The positive cross-price derivative $\frac{\partial M_d}{\partial P_a}$ captures the vertical linkage: a rise in the alfalfa price increases dairy production costs, which contracts domestic dairy supply and subsequently

drives up the demand for dairy imports. ROW dairy export supply is $X_d(P_d^*)$, with $X_d' > 0$.

To formalize the cost pass through mechanism, We model dairy production with a Cobb–Douglas cost function where alfalfa represents a cost share $\alpha \in (0, 1)$:

$$MC_d = A P_a^\alpha w^{1-\alpha}, \quad (3)$$

where w denotes the composite price of other inputs (e.g., labor, silage, capital). Under this specification, a one percent increase in the alfalfa price raises dairy marginal cost by α percent. This provides the fundamental mechanism through which upstream tariff shocks transmit to the downstream sector.

We define the following constant elasticities for the log-linear specification:

- $\eta_a < 0$: elasticity of China’s alfalfa import demand with respect to P_a ;
- $\sigma_a > 0$: elasticity of ROW alfalfa export supply with respect to P_a^* ;
- $\eta_d < 0$: elasticity of China’s dairy import demand with respect to P_d ;
- $\sigma_d > 0$: elasticity of ROW dairy export supply with respect to P_d^* ;
- $\phi > 0$: cross-elasticity of dairy import demand with respect to P_a , reflecting the vertical linkage. Under the cost structure in equation (3), ϕ is proportional to the alfalfa cost share α and the domestic dairy supply elasticity.

The two market-equilibrium conditions are:

$$M_a(P_a) = X_a(P_a^*), \quad (4)$$

$$M_d(P_d, P_a) = X_d(P_d^*). \quad (5)$$

3.2 Comparative statics: twin tariff effects

To analyze the magnitude of price transmission across the supply chain, We log-linearize the equilibrium conditions around the free-trade benchmark. Let $\hat{\tau}_j \equiv d \ln(1 + \tau_j)$ represent the

change in the tariff for good $j \in \{a, d\}$.

Alfalfa market: Log-differentiating the alfalfa equilibrium condition in (4) with the tariff wedge (1) yields:

$$\hat{P}_a = \frac{\sigma_a}{\sigma_a - \eta_a} \hat{\tau}_a, \quad \hat{P}_a^* = \frac{\eta_a}{\sigma_a - \eta_a} \hat{\tau}_a. \quad (6)$$

Given that $\sigma_a > 0$ and $\eta_a < 0$, the domestic alfalfa price rises by a fraction $\sigma_a/(\sigma_a - \eta_a) \in (0, 1)$ of the tariff, while the world price falls. This downward pressure on the world price represents the terms-of-trade effect, while foreign exporters absorb a portion of the tariff burden (Schmitz et al., 2022; Schaefer and Wolf, 2025).

Proposition 1 (Input tariff effects). *When China imposes an alfalfa tariff τ_a , the domestic price rises by less than the full tariff amount due to a terms-of-trade improvement $|\eta_a|/(\sigma_a - \eta_a) > 0$. The domestic price pass-through rate, $\sigma_a/(\sigma_a - \eta_a)$, is increasing in the elasticity of foreign export supply (σ_a) and decreasing in the elasticity of China's import demand ($|\eta_a|$).*

Dairy market: To determine the impact on the downstream dairy sector, we log-differentiate equation (5) and substitute the results from the alfalfa market equation (6):

$$\hat{P}_d = \underbrace{\frac{\sigma_d}{\sigma_d - \eta_d} \hat{\tau}_d}_{\text{Direct dairy tariff effect}} + \underbrace{\frac{\phi \sigma_a}{(\sigma_d - \eta_d)(\sigma_a - \eta_a)} \hat{\tau}_a}_{\text{Cross-market (twin tariff) channel}}. \quad (7)$$

Equation (7) represents the central structure result of the model. The domestic dairy price is a function of both the direct output tariff and the upstream input tariff. The second term—the twin tariff channel—shows that the alfalfa tariff transmits downstream with a magnitude dictated by the strength of the vertical linkage (ϕ), the alfalfa pass-through rate ($\sigma_a/(\sigma_a - \eta_a)$), and the structural elasticity of the dairy market ($1/(\sigma_d - \eta_d)$).

Proposition 2 (Cross-Market Pass-Through). *An upstream alfalfa tariff raises the domestic dairy price through the vertical cost linkage. The cross-market elasticity of the domestic dairy*

price with respect to the alfalfa tariff is

$$\frac{\partial \hat{P}_d}{\partial \hat{\tau}_a} = \frac{\phi \sigma_a}{(\sigma_d - \eta_d)(\sigma_a - \eta_a)} > 0.$$

This effect is increasing in the alfalfa cost share α (through ϕ) and the foreign alfalfa export supply elasticity σ_a , while being attenuated by higher dairy import demand elasticity $|\eta_d|$.

While Proposition 2 characterizes the equilibrium price response, the total economic burden of the twin-tariff regime is also a function of quantity adjustments. Because the upstream and downstream markets exhibit different degrees of substitutability and supply elasticities, the twin-tariff shock creates a "wedge" that distorts trade volumes and social surplus asymmetrically, which leads the following proposition:

Proposition 3 (Asymmetric Import Substitution). *When the ROW dairy export supply is more elastic than the alfalfa export supply ($\sigma_d > \sigma_a$), dairy import volumes respond more to tariffs than alfalfa imports. Despite this, the alfalfa tariff may generate a disproportionate DWL per dollar of revenue because its distortionary effects are amplified downstream through the dairy supply chain.*

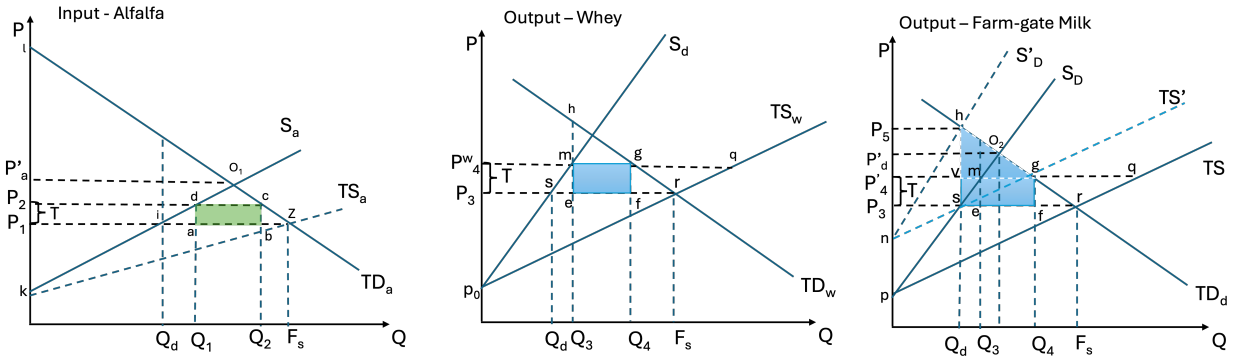


Figure 2: Twin tariffs: simultaneous shocks to input and output markets

Figure 2 provides a graphical illustration of the "twin-tariff" mechanism, and visualizes the structural price transmission described in Proposition 2 to the asymmetric welfare distortions highlighted in Proposition 3. The left panel depicts the alfalfa market, where the imposition

of a tariff T creates a price wedge, raising the domestic price from P_1 to P_2 . This price increase induces a contraction in import quantity from Q_2 to Q_1 , represented by the green shaded area $abcd$. The whey is imported mainly from the United States, so the twin tariffs can shift the domestic farm-gate milk supply curve through cost linkage and increase the price of whey. Because the domestic supply of high-quality alfalfa is limited and China acts as a large-country importer, this upstream distortion is the primary catalyst for the downstream effects. The resulting price hike in the forage market represents a direct increase in the marginal cost of production for the dairy sector.

The connection between the two panels illustrates the vertical linkage described in Proposition 2. As the domestic alfalfa price rises to P_2 , the domestic dairy supply curve in the right panel shifts leftward from S_D to S'_D . This shift represents the cost pass-through, where the upstream tariff burden is transmitted downstream to the dairy sector. Even in the absence of a direct dairy tariff, this cross-market transmission would drive the milk price from P_3 up toward P'_d , as the intersection of S'_D and the total demand TD_d moves upward. The right panel further illustrates the compounding effect when a direct dairy tariff T (shifting the price from P_3 to P_4) is applied simultaneously with the input cost shock. The final equilibrium price reaches P_5 , a level significantly higher than if either tariff had operated in isolation. The blue shaded area $efgh$ and the associated triangular regions represent the compounded DWL.

3.3 Welfare impacts of twin-tariff

We define total welfare in China's dairy-feed system as the sum of consumer surplus, producer surplus, and government tariff revenue to quantify the welfare impacts of the twin-tariff regime:

$$W = CS_d(P_d) + PS_a(P_a) + PS_d(P_d, P_a) + TR_a + TR_d, \quad (8)$$

where $TR_j = \tau_j P_j^* M_j$ represents the tariff revenue from good $j \in \{a, d\}$.

The net welfare change under the twin-tariff regime relative to a free trade benchmark can be decomposed into three primary components:

$$\Delta W = \underbrace{(P_a^{*ft} - P_a^*)M_a + (P_d^{*ft} - P_d^*)M_d}_{\text{Terms-of-trade gains}} - \underbrace{DWL_a - DWL_d}_{\text{Dead weight losses}} - \underbrace{\Delta C_{a \rightarrow d}}_{\text{Cross-market distortion}}, \quad (9)$$

where $DWL_j = \frac{1}{2} \tau_j \Delta P_j |\Delta M_j|$ is the Harberger triangle in market j , and the cross-market distortion is:

$$\Delta C_{a \rightarrow d} = (P_a - P_a^{ft}) \cdot \bar{\alpha} \cdot \bar{Q}_d^s. \quad (10)$$

This term captures the additional cost burden imposed on dairy producers by the alfalfa tariff. It is defined as the tariff-induced alfalfa price increase weighted by the cost share (α) and total domestic dairy output. $\bar{\alpha} = \alpha \bar{P}_d / \bar{P}_a$. This component is unique to the twin-tariff setting; it represents the welfare loss that propagates through the supply chain, a factor often omitted in single-commodity trade analysis (Amiti et al., 2019; Fajgelbaum and Khandelwal, 2022; Staples and McCullough, 2025).

Proposition 4 (Twin Tariff Amplification). *The welfare loss from twin tariffs can exceed the sum of the losses from isolated input and output tariffs when the cross-market cost distortion is sufficiently large relative to any offsetting tariff-revenue or terms-of-trade gains.*

$$|\Delta W(\tau_a, \tau_d)| > |\Delta W(\tau_a, 0)| + |\Delta W(0, \tau_d)|.$$

This super-additivity arises from the cross-market distortion $\Delta C_{a \rightarrow d}$. The alfalfa tariff shifts the domestic dairy supply curve inward, which endogenously amplifies the DWL generated by the subsequent dairy tariff.

3.4 Multi-origin extension of the dairy market

The benchmark model treats the rest of the world as a composite dairy exporter. To connect the model more directly to the empirical setting, in which Chinese importers can substitute U.S.-origin dairy with New Zealand and EU origins, we extend the dairy export-supply side. Let total dairy export supply available to China be

$$X_d(P_d^*) = X_d^{US}(P_d^{US}) + X_d^{NZ/EU}(P_d^{NZ/EU}), \quad (11)$$

where X_d^{US} denotes dairy supply from the United States and $X_d^{NZ/EU}$ denotes supply from alternative suppliers (mainly New Zealand and the European Union). The dairy tariff applies to U.S.-origin imports but not to alternative origins, so the relative delivered prices become

$$P_d^{US, \text{tariff}} = (1 + \tau_d) P_d^{US, *}, \quad P_d^{NZ/EU, \text{tariff}} = P_d^{NZ/EU, *}. \quad (12)$$

When the U.S. delivered price rises, Chinese importers substitute toward NZ/EU dairy. We define the period- t substitution share as

$$s_{\text{sub}, t} = \min \left\{ \frac{\Delta M_{d,t}^{NZ/EU}}{|\Delta M_{d,t}^{US}|}, 1 \right\}, \quad (13)$$

with the upper bound of unity reflecting the institutional constraint that NZ/EU substitution can offset, but not exceed, the displaced U.S.-origin volume. The substitution-adjusted dairy deadweight loss is then

$$DWL_{d,t}^{\text{final}} = DWL_{d,t}^{\text{raw}} (1 - s_{\text{sub}, t}) + DWL_{d,t}^{\text{residual}}, \quad (14)$$

where $DWL_{d,t}^{\text{raw}}$ is the dairy DWL implied by a single-origin model (Equation (9)) and the residual term captures the remaining welfare cost when alternative-origin dairy replaces U.S.-origin dairy at a higher effective price:

$$DWL_{d,t}^{\text{residual}} = \frac{1}{2} \left(P_d^{NZ/EU} - P_d^{US,ft} \right) \left| \Delta M_{d,t}^{\text{sub}} \right|. \quad (15)$$

The residual is positive whenever NZ/EU dairy enters at a higher effective price than the free-trade U.S.-origin price. Empirically this is generally the case for whey, the main U.S.-origin dairy line, which has a unit-value advantage relative to NZ/EU alternatives. Equation (14) therefore distinguishes pure origin reallocation, which carries no welfare cost beyond the initial tariff distortion, from the residual price-premium distortion associated with sourcing the displaced quantity at a higher effective price.

3.5 The impacts of domestic producer subsidy policy

Beginning in 2020, China introduced producer subsidies for domestic forage growers to mitigate the cost shocks stemming from the alfalfa tariff.

Proposition 5 (Subsidy Offset). *Under a twin-tariff regime (τ_a, τ_d) , a domestic alfalfa production subsidy (s_a) can fully offset the cross-market distortion in Equation (10) if:*

$$s_a \geq \alpha \tau_a P_a^*.$$

While the optimal subsidy effectively eliminates the vertical cost propagation channel, it cannot resolve the direct (DWL_d) caused by the dairy output tariff. The net welfare loss under this subsidy is reduced to:

$$\Delta W^{\text{sub}} = \left(P_d^{\text{ft}} - P_d^* \right) M_d - DWL_d - G(s_a),$$

where $G(s_a)$ is the fiscal cost of the subsidy. This intervention is welfare-improving relative to the twin-tariff baseline whenever the avoided cross-market distortion exceeds the fiscal cost.

Figure 3 illustrates the subsidy offset mechanism discussed in Proposition 5, showing how a domestic intervention in the upstream market can decouple the vertically linked "twin-tariff"

distortion. The introduction of a input (alfalfa) producer subsidy s_a shifts the domestic alfalfa supply curve outward from S_a to S'_a (lowering the intercept from k to k'). This shift effectively neutralizes the tariff-induced price hike, bringing the market-clearing price back down toward P_1 . A subsidy equivalent to the area $P_1P'_2zc$ would stabilize the alfalfa price and prevent the supply curves for whey and farm-gate milk from shifting. Through the cost linkage, the dairy supply curve shifts back toward its free-trade position, partially reversing the twin tariff distortion (Kondaridze and Luckstead, 2023).

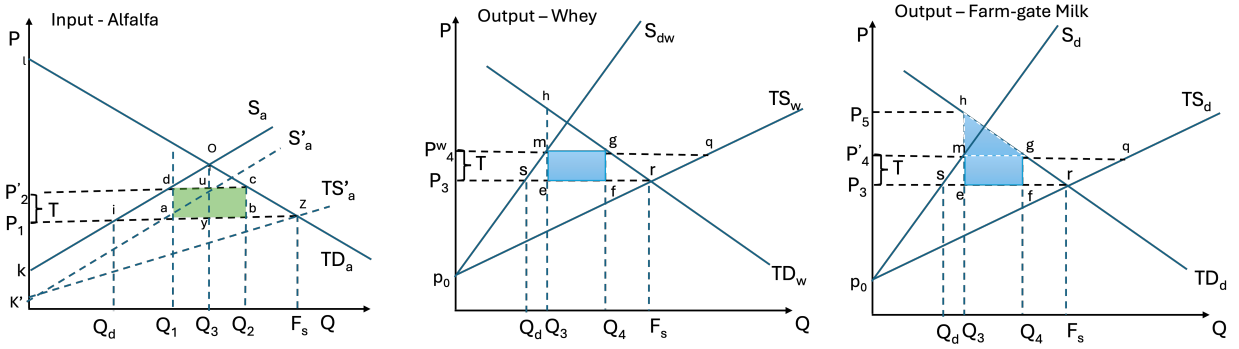


Figure 3: Domestic alfalfa subsidy as countermeasure to twin tariffs

4 Data and Descriptive Insights

4.1 Data sources and variable construction

We construct a monthly time-series dataset spanning January 2005 through August 2025.

Table 1 summarises the data sources, codes, and time coverage of each series.

Table 1: Data sources, codes, and time coverage of variables used in the empirical analysis.

Variable	Source	Code	Unit	Time
Q_D^{US} : U.S. dairy export qty to China	USDA FAS (GATS)	HS 04, 0406	Metric ton	01/2005–06/2025
Q_D^{US} value (for unit-value P_d^{US})	USDA FAS (GATS)	HS 04, 0406	\$ thousand	01/2005–06/2025
Q_A^{US} : U.S. alfalfa export qty to China	USDA FAS (GATS)	HS 1214	Metric ton	01/2005–06/2025
P^{alf} value (for unit-value)	USDA FAS (GATS)	HS 1214	\$ thousand	01/2005–06/2025
P^{milk} : China farm-gate milk price	CLAL.it (NBS aggregator)	—	CNY / kg	01/2009–06/2025
τ^A, τ^D : applied tariff rates	GACC; Bown (2023)	—	Ad valorem (%)	06/2018–03/2025
NZ/EU dairy import quantities	UN Comtrade / GACC	HS 04, 0406	Metric ton	01/2010–06/2024
China forage-grass imports (robustness)	GACC	HS 12	Metric ton	01/2010–12/2024
FAO dairy price index	FAO (avg. 2014–2016 = 100)	HS 04, 0410	Index	01/2005–07/2025
Brent: international transport proxy	U.S. EIA	—	\$ / barrel	01/2005–06/2025
COVID: COVID-19 dummy	—	—	0/1 indicator	03/2020–12/2021

- *Trade flows*: Monthly U.S. export quantities and values to China for alfalfa hay (HS 1214) and dairy products (HS 04) are obtained from the USDA Foreign Agricultural Service Global Agricultural Trade System (GATS). We calculate unit import values (value-to-quantity ratios) as proxies for bilateral trade prices. To account for broader market trends, we supplement these with China’s aggregate forage-grass imports from China Customs (GACC).
- *Farmgate Milk Prices*: China’s monthly farm-gate milk price (CNY/kg) is sourced from CLAL.it, which aggregates National Bureau of Statistics (NBS) data into a nationally uniform series. The use of a national price is justified by the high concentration of

the Chinese processing sector—dominated by Mengniu and Yili—which reduces inter-regional price dispersion (Zhang and Abler, 2023). We also include the FAO Dairy Price Index to control for global price volatility.

- *Alfalfa Prices*: the CIF unit value of U.S.-origin alfalfa shipped to China (constructed as the value to-quantity ratio from the GATS series)
- *Tariff Rates*: Monthly applied tariff rates for alfalfa and representative dairy categories (whey and milk powder) are compiled from GACC and the Bown (2023) database. Following Xu et al. (2025), we focus on these primary lines as they constitute the bulk of U.S. dairy exports to China, excluding low-volume, high-variation lines.

4.2 The twin-tariff regime and policy windows

The empirical analysis exploits the variation generated by China’s retaliatory tariff schedule against U.S. agricultural goods between 2018 and 2025. Two features of this regime are central to the interpretation of the regression results: the simultaneous imposition of duties on an upstream input (alfalfa) and a downstream output (dairy)—the “twin” structure—and the discrete policy phases that allow us to compare effects across windows.

Policy chronology. In July 2018, China imposed an additional 25% retaliatory tariff on U.S. agricultural goods including alfalfa, while most U.S. dairy products became subject to substantial retaliatory duties (Xu et al., 2025). By 2019, many cheese lines faced totals around 33–37%, reflecting the MFN tariff plus the 25% Section 301 retaliation. China announced an additional round in September 2019, adding 5–10% on selected U.S. goods. After the Phase One Agreement entered into force on February 14, 2020, China did not fully remove retaliatory tariffs but opened an exclusion process for affected farm goods; whey for feed use received repeated exclusions through February 2025, and alfalfa benefited from exclusion treatment beginning in September 2019. In May 2025, following the Geneva negotiations, China temporarily reduced the additional retaliatory tariff imposed during the 2025 escalation

from 125% to 10% for 90 days. Figure A5 maps these policy events onto the four windows used in the analysis.

Four policy windows. The chronology defines the four windows used in the period-level interrupted-time-series specification (21) and in the welfare counterfactual:

- *Control* (2017–July 2018): pre-tariff baseline.
- *First Trade War* (July 2018–2022): imposition and entrenchment of retaliatory duties; the period during which Phase One exclusions partially eased the effective rate.
- *Adjustment* (2022–2023): tariff exclusions remain active and domestic alfalfa subsidies are in force; importers and processors begin systematic origin diversification.
- *Second Trade War* (2024–2025): renewed escalation in 2025 reaching 125%, partially reversed by the May 2025 Geneva agreement; the period during which adaptation channels are most active.

4.3 Descriptive insights

Tariff Rates. Figure 4 shows the 2005–2025 time series of τ_t^A and τ_t^D . The series are flat at zero through mid-2018, jump in July 2018 (alfalfa) and during 2019 (dairy), and remain elevated thereafter with a brief spike in 2025. This sustained step structure is the variation that identifies the tariff coefficients in Equations (16)–(18) and provides the relevance condition for the 2SLS estimator.

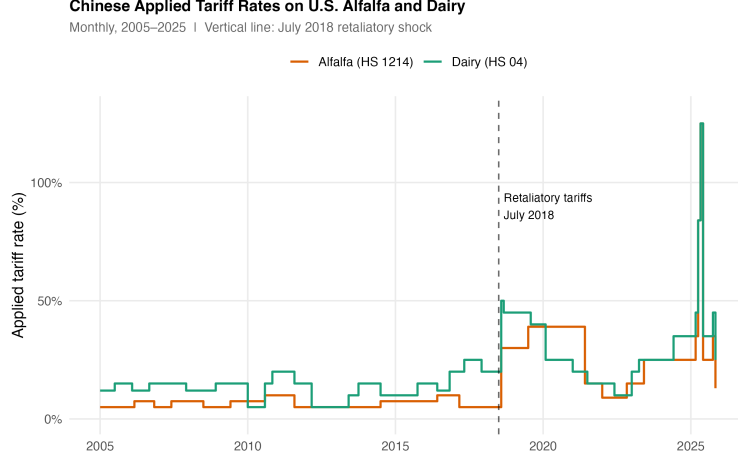
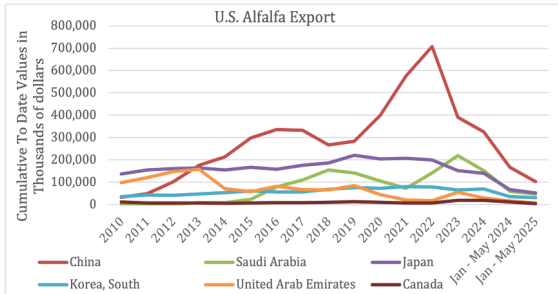
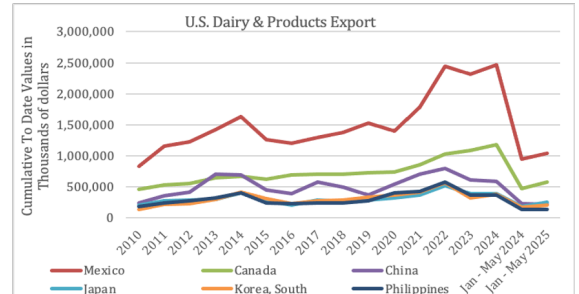


Figure 4: Time series of the treatment variables τ_t^A and τ_t^D : Chinese applied tariff rates on U.S. alfalfa and dairy products, 2005–2025. The pre-2018 zero baseline and the post-2018 step structure provide the identifying variation for the regression analysis. Sources: China Customs (GACC); Bown (2023).

Trade-flow structural break. Figure 5 reports U.S. export values to China for alfalfa hay (panel a) and dairy products (panel b) from 2010 through 2025. Both series exhibit a clear structural break in 2018–2019: alfalfa export value contracts sharply after July 2018, and dairy export value contracts after the May 2019 escalation, with both series stabilising at substantially lower levels through 2022. The second trade war (2023–2025) produces a further contraction in alfalfa exports but a more muted dairy decline—an asymmetry consistent with the dynamic adaptation finding in Section 6.2. The post-2018 shock is the natural identifying variation for Equation (16).



(a) U.S. alfalfa hay exports (HS 1214)



(b) U.S. dairy exports (HS 04)

Figure 5: Trade-flow context: U.S. exports to China, 2010–2025. Both series exhibit clear structural breaks corresponding to the July 2018 alfalfa tariff and the 2019 dairy escalation. Source: U.S. Census Bureau Trade Data.

Asymmetric quantity-price response. Figure 6 brings the import-quantity and unit-price dimensions together for both commodities. Three features stand out. First, the post-2018 contraction in alfalfa import quantities is large and persistent, while the alfalfa unit value rises sharply—visualising the inelastic-supply prediction underlying Proposition 3: the alfalfa tariff burden falls on price more than on quantity. Second, the dairy quantity adjustment is sharper than the dairy price adjustment, consistent with origin substitution toward NZ/EU absorbing the displaced volume rather than the tariff being passed through. Third, the cross-panel timing is offset: alfalfa volumes break in 2018Q3, dairy volumes break in 2019Q2, mirroring the policy timeline in Figure A5. The dependent variables in Equations (16) and (18) are the differenced log versions of these series.

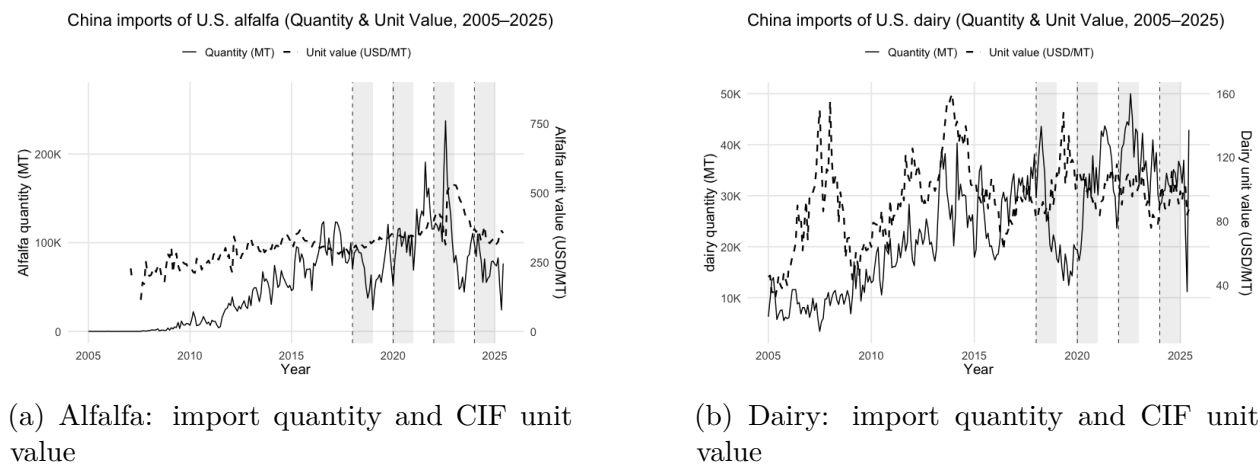


Figure 6: Joint quantity-price dynamics for Chinese U.S.-origin alfalfa (left) and dairy (right) imports, 2010–2025. Sources: GACC; USDA-FAS-GATS.

Domestic price evolution and producer-margin compression. Figure 7a reports the CIF unit value of U.S.-origin alfalfa shipped to China; the post-2018 level is visibly higher than the pre-tariff baseline, consistent with substantial pass-through to Chinese feed costs. Figure 7b reports the CLAL.it national farm-gate milk price (CNY/kg) from 2016–2025, showing seasonal cycles and an upward trend during 2018–2021. Figure 7c ties the two series together: it indexes both to a common base and reveals that alfalfa feed costs rose proportionately more than the farm-gate milk price during 2018–2022, a producer-margin compression that

the period-effects analysis quantifies in the results.

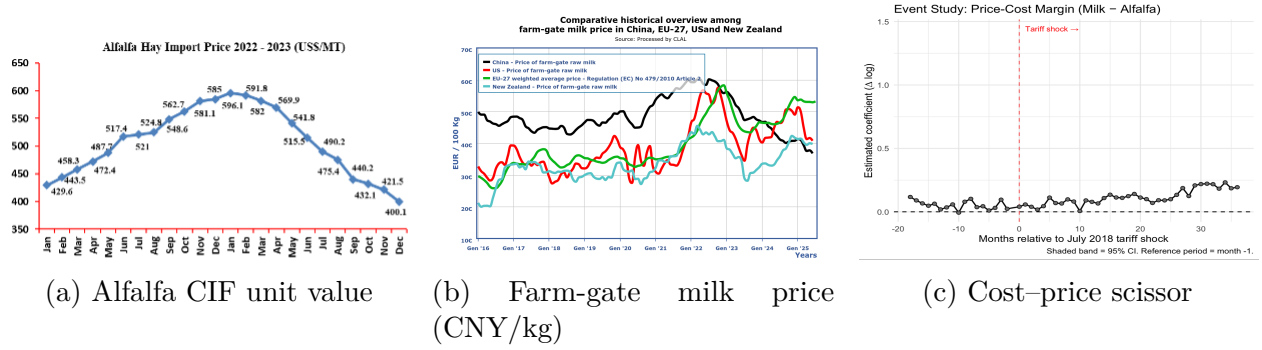


Figure 7: Domestic price evolution. The alfalfa CIF unit value (left) rises sharply post-2018; the farm-gate milk price (centre) shows seasonal cycles and an upward trend through 2021; the indexed cost-price scissor (right) shows that alfalfa cost rose proportionately more than the farm-gate milk price during 2018–2022, indicating producer-margin compression. Sources: U.S. Department of Agriculture, Foreign Agricultural Service (USDA FAS) (2021); CLAL.it.

Summary statistics by policy window. Table 2 summaries the means of the key time-varying variables by policy window. The qualitative patterns above are quantitatively reflected in the table: U.S. alfalfa import quantities fall sharply across windows; U.S. dairy import quantities also fall but by a smaller proportion (with NZ/EU substitution absorbing the gap); the alfalfa CIF price rises substantially during the First Trade War; and the farm-gate milk price rises by a smaller proportion—the producer-margin compression visualised in Figure 7c.

The descriptive evidence above motivates each piece of the four-equation strategy in Section 5. The structural break in trade flows (Figure 5) is the source of identifying variation for Equation (16). The asymmetric quantity-price response (Figure 6) and the contrast between concentrated alfalfa origin and diversified dairy origin (Appendix Figures A1–A4) provide the empirical foundation for Proposition 3. The visible co-movement of alfalfa CIF and farm-gate milk prices (Figure 7) motivates the reduced-form pass-through specification of Equation (18). The cost-price scissor previews the producer-margin compression analysed in the period-effects subsection.

Table 2: Summary statistics by policy window. Means of monthly observations.

Variable	Control (2017- 2018)	1st Trade War (2018– 2022)	Adjustment (2022- 2023)	2nd Trade War (2023- 2025)
Alfalfa import from U.S. (1k T/Mo)	94.0	65.3	73.9	70.1
Dairy import from U.S. (1k T/Mo)	35.8	18.8	35.2	31.7
Dairy import alternative (1k T/Mo)	154.4	166.3	157.0	145.2
Alfalfa CIF unit value (USD/ton)	290.8	328.9	446.7	350.3
Farm-gate milk price (CNY/kg)	3.47	3.88	3.76	3.20
Tariff rate, alfalfa (% , period end)	5	25	25	13
Tariff rate, dairy (% , period end)	20	39	25	25
Months in window	18	42	12	29

Notes: Values are monthly means by policy window. U.S. alfalfa quantity, U.S. dairy quantity, alfalfa CIF unit value, and farm-gate milk price are computed from the monthly trade and price dataset. NZ/EU dairy import quantity is computed from the monthly China dairy-import-by-origin series for HS 0401–0406, aggregating New Zealand and EU-27 origins. The Second Trade War window uses available observations through November 2025 for U.S.-origin and price variables; the NZ/EU row uses available Comtrade-origin observations through 2024. The end-of-window tariff rates are representative retaliatory ad valorem rates; effective rates vary across HS6 lines and exclusion periods.

5 Empirical Strategy

5.1 Main specification

We estimate four model specifications that progressively examine the linkage between tariff shocks, trade flows, and domestic prices, with a specific focus on the potential "additionality" of twin tariffs.

First, we document how tariff changes drive the import volumes from the United States. This serves as a direct test of the cost-pass-through margin and functions as the first-stage diagnostic for the two-stage least squares (2SLS) model estimated later. Second, we estimate the reduced-form association between tariff changes and farm-gate milk prices, capturing the net effects of feed-cost shocks and trade-diversion. Third, we add a contemporaneous interaction term to the reduced form analysis to test whether the joint twin-tariff configuration

alters the price response beyond the additive sum of the individual elasticities. Last, we employ a 2SLS framework to isolate the import-volume channel from the broader reduced-form association.

Mapping to the conceptual model. The four specifications correspond to distinct elements of the conceptual model in Section 4, and several propositions are tested elsewhere through complementary specifications. Equation (16) provides auxiliary documentation of the trade-flow response and the first-stage relevance check for the 2SLS. Equation (18) estimates the within-year empirical analogue of the cross-market price pass-through in Proposition 2: the cumulative coefficient $\sum_k \hat{\theta}_k^A$ is the empirical counterpart of $\partial \hat{P}_d / \partial \hat{\tau}_a = \phi \sigma_a / [(\sigma_d - \eta_d)(\sigma_a - \eta_a)]$ from Equation (7). Equation (19) tests the additive log-linear structure that underlies Proposition 2’s closed-form pass-through expression; a non-zero $\hat{\theta}_{AD}$ rejects this additivity and signals second-order curvature not captured by the first-order conceptual model. Equation (20) attempts to isolate the import-volume channel from the broader reduced-form association; this is a structural decomposition exercise rather than a test of any single proposition. The welfare super-additivity of Proposition 4 and the subsidy offset of Proposition 5 are tested through the calibrated counterfactual analysis in Section 6, and the alfalfa-tariff incidence prediction of Proposition 1 is reflected in the alfalfa CIF-price response reported in Table 2.

Before evaluating domestic price effects, we first document that tariffs significantly alter trade flows. This confirms the economic relevance of the tariff variation and tests the asymmetric substitutability prediction of Proposition 3. We estimate the following equation:

$$\Delta \ln Q_{D,t} = \alpha_0 + \sum_{k=0}^2 \left(\gamma_k^A \Delta \ln(\text{tr}_{t-k}^A + 1) + \gamma_k^D \Delta \ln(\text{tr}_{t-k}^D + 1) \right) + \gamma' \mathbf{X}_t + \mu_m + \lambda_y + u_t, \quad (16)$$

where $\Delta \ln Q_{D,t}$ is the monthly log-difference of China’s dairy import quantity from the United States. The coefficients γ_k^A and γ_k^D measure the tariff–volume elasticity at horizon k for alfalfa and dairy tariffs, respectively. Both tariffs are included because they may simultaneously shift the import volume: the dairy tariff directly suppresses import demand by raising the

relative price of U.S. dairy relative to alternative suppliers, such as New Zealand and EU; and the alfalfa tariff operates indirectly through the input-cost channel, where higher feed costs compress producer margins and reduce derived demand for imported dairy intermediates such as whey and milk powder. $\mathbf{X}_t$ is a vector of control variables, which include: Brent_t , The monthly Brent crude oil price index (U.S. EIA) serves as a proxy for international transport-cost variation; COVID_t , A binary indicator for the 2020–2021 period is included to absorb global supply-chain disruptions; and FAO Dairy Price Index, which services as control for the international dairy demand shocks.

We interpret Equation (16) as an auxiliary test of the cross-tariff sign prediction implicit in Proposition 2: a negative cumulative $\sum_k \gamma_k^A$ is consistent with the upstream–downstream cost-pass-through channel, though this specification does not separate the derived demand component from origin-reallocation.

The within-equation comparison in Equation (16) is distinct from the across-market quantity prediction of Proposition 3, which compares the cumulative dairy-import response to the cumulative alfalfa-import response. To evaluate that prediction, we estimate Equation (16) in parallel with $\Delta \ln Q_{A,t}$ (China’s U.S.-origin alfalfa import quantity) as the dependent variable. Because the Prop. 3 comparison is across markets rather than within a single equation, we report it using a pooled twin-tariff regressor, $\ln(1 + \tau_t^A) + \ln(1 + \tau_t^D)$, which yields a single comparable cumulative coefficient for each outcome; the corresponding unrestricted estimates for U.S.-origin dairy imports appear in Column (1) of Table 4. The full distributed-lag estimates appear in Tables 9 and 11.

Block-recursion check. The conceptual model in Section 4 treats the alfalfa market as upstream of the dairy market: alfalfa-market clearing depends on the alfalfa tariff and ROW supply but not on conditions in the dairy market, while dairy-market clearing depends on both prices through the cross-derivative $\partial M_d / \partial P_a > 0$ (Equation (5)). This block-recursive ordering is an identifying assumption: it permits using the alfalfa tariff as an instrument that affects dairy import volumes only through the input-cost channel, with no contamination

from a reverse downstream-to-upstream effect. We test this assumption directly by estimating an auxiliary regression of U.S.-origin alfalfa imports on both tariff series:

$$\Delta \ln Q_{A,t} = \rho_0 + \sum_{k=0}^2 \left[\rho_k^A \Delta \ln(1 + \tau_{t-k}^A) + \rho_k^D \Delta \ln(1 + \tau_{t-k}^D) \right] + \boldsymbol{\rho}' \mathbf{X}_t + \mu_m + \lambda_y + \varepsilon_t. \quad (17)$$

Under block recursion, the dairy-tariff coefficients $\{\rho_k^D\}_{k=0}^2$ should be small in magnitude and jointly indistinguishable from zero. Table 3 reports the estimates: each individual dairy-tariff coefficient is statistically insignificant ($p > 0.29$ at every horizon), and the joint Wald test fails to reject $\rho_0^D = \rho_1^D = \rho_2^D = 0$ ($F = 1.188$, $p = 0.315$). We interpret this as evidence against reverse downstream-to-upstream feedback, supporting the block-recursive ordering used in both the conceptual model and the 2SLS specification in Equation (20). The alfalfa-tariff coefficients are also imprecisely estimated in this auxiliary regression, so Table 3 should be read as a falsification check on dairy-tariff feedback rather than as a structural estimate of the alfalfa-import demand response.

Table 3: Block-recursion check: alfalfa imports and dairy tariffs

Variable	Coefficient	Std. error	<i>p</i> -value
$\Delta \ln(1 + \tau_t^A)$	0.904	2.240	0.688
$\Delta \ln(1 + \tau_{t-1}^A)$	-2.325	2.480	0.349
$\Delta \ln(1 + \tau_{t-2}^A)$	-0.678	2.030	0.738
$\Delta \ln(1 + \tau_t^D)$	-1.968	2.580	0.446
$\Delta \ln(1 + \tau_{t-1}^D)$	2.961	2.840	0.298
$\Delta \ln(1 + \tau_{t-2}^D)$	0.617	2.010	0.759
Joint test: dairy-tariff lags = 0	$F = 1.188$		$p = 0.315$
Month and year fixed effects, controls	Yes		
Observations	245		

Notes: The dependent variable is $\Delta \ln Q_{A,t}$, the monthly log change in China's U.S.-origin alfalfa import quantity. The model includes contemporaneous, one-month-lagged, and two-month-lagged changes in alfalfa and dairy tariff rates, macro controls (Brent crude, COVID-19 dummy), and month and year fixed effects. Heteroskedasticity-robust standard errors are reported. The joint Wald test on the dairy-tariff coefficients tests the block-recursion null $\rho_0^D = \rho_1^D = \rho_2^D = 0$.

Having established that tariffs meaningfully shift trade flows, we next examine whether tariff changes co-move with domestic farm-gate milk prices using the following reduced-form

specification:

$$\Delta \ln P_t^{\text{milk}} = \theta_0 + \sum_{k=0}^2 \left(\theta_k^A \Delta \ln(\text{tr}_{t-k}^A + 1) + \theta_k^D \Delta \ln(\text{tr}_{t-k}^D + 1) \right) + \boldsymbol{\theta}' \mathbf{X}_t + \mu_m + \lambda_y + \nu_t, \quad (18)$$

where $\Delta \ln P_t^{\text{milk}}$ is the monthly log-difference of China's farm-gate milk price.² The coefficients θ_k^A and θ_k^D measure the within-year tariff-price association at horizon k . We include up to two monthly lags ($k \in \{0, 1, 2\}$) to account for typical shipping durations and contracting delays in international agricultural trade (Chor and Li, 2024; Kim and Steinbach, 2023). Thus, the cumulative sums $\sum_k \hat{\theta}_k^A$ and $\sum_k \hat{\theta}_k^D$ capture the three-month impulse response. A positive cumulative alfalfa coefficient is consistent with feed-cost pass-through. In contrast, the dairy-tariff coefficient is not sign-restricted, as origin substitution or precautionary stockpiling may offset the standard positive association at a monthly frequency.

To test whether co-occurring tariff shocks generate a price response that deviates from the sum of independent shocks, we add a contemporaneous interaction term to Equation (18)

$$\begin{aligned} \Delta \ln P_t^{\text{milk}} = & \theta_0 + \sum_{k=0}^2 \left[\theta_k^A \Delta \ln(1 + \text{tr}_{t-k}^A) + \theta_k^D \Delta \ln(1 + \text{tr}_{t-k}^D) \right] \\ & + \theta_{AD} \left[\Delta \ln(1 + \text{tr}_t^A) \times \Delta \ln(1 + \text{tr}_t^D) \right] \\ & + \boldsymbol{\theta}' \mathbf{X}_t + \mu_m + \lambda_y + \nu_t \end{aligned} \quad (19)$$

The coefficient θ_{AD} captures the contemporaneous deviation from additivity. A positive $\hat{\theta}_{AD}$ suggests super-additive price amplification, where joint shocks raise prices by more than the individual effects combined—consistent with reinforced cost pressure on both upstream and downstream margins. Conversely, a negative $\hat{\theta}_{AD}$ indicates sub-additive price compression, suggesting that parallel adaptation responses (e.g., rapid inventory release or contract renegotiation) may dampen the net price signal.

The reduced form in Equation (18) captures the total within-year tariff-price association

²We use CNY/kg as our unit of measurement. We apply a +1 additive shift to tariff rates before taking logs to ensure the transformation is defined during zero-tariff periods.

and the interaction in Equation (19) probes the additive structure underlying Proposition 2, but neither isolates the contribution of any single transmission channel. To isolate the import-volume channel from other transmission mechanisms, we estimate a 2SLS specification where the tariff series instrument for U.S.-origin dairy import volumes. The estimating equation for the second stage is:

$$\Delta \ln P_t^{\text{milk}} = \beta_0 + \beta_Q \widehat{\Delta \ln Q_{D,t}} + \beta' \mathbf{X}_t + \mu_m + \lambda_y + \epsilon_t, \quad (20)$$

where $\widehat{\Delta \ln Q_{D,t}}$ is the fitted value from Equation (16). This isolates the exogenous, tariff-driven component of dairy import volume changes. To satisfy the exclusion restriction, we include contemporaneous and lagged alfalfa CIF prices ($\Delta \ln P_{t-k}^{\text{alf}}$) as exogenous controls in the second stage. This is intended to absorb the direct feed-cost channel through which the alfalfa tariff affects milk prices independently of dairy import volumes.

Our analysis relies on two primary assumptions. First, we treat the tariff series as predetermined with respect to short-run domestic dairy market conditions. China’s retaliatory schedule was largely dictated by geopolitical negotiations and Section 301 sequencing rather than contemporaneous milk-market shocks (Amiti et al., 2019; Xu et al., 2025). Supplemental Robustness Checks Summary in Appendix support this assumption. Second, we assume the exclusion restriction holds after controlling for alfalfa CIF prices. While including these prices introduces potential "bad-control" concerns—as CIF prices are post-treatment mediators—we mitigate this through the inclusion of fuel-price controls and fixed effects to absorb common unobserved drivers of commodity price co-movement.

5.2 Period-level analysis

We estimate the period-level shifts in the farm-gate milk price using an interrupted time-series specification:

$$\ln P_t^{\text{milk}} = \alpha + \beta t + \sum_{p \in \{\text{TW1}, \text{Adj}, \text{TW2}\}} \delta_p \mathbf{1}\{t \in P_p\} + \gamma' \mathbf{X}_t + \mu_m + \varepsilon_t, \quad (21)$$

where the indicators $\mathbf{1}\{t \in P_p\}$ flag the first trade war (TW1, 2018–2022), adjustment period (Adj, 2022–2023), and second trade war (TW2, 2023–2025), with the 2017–2018 pre-tariff window as the omitted baseline. The coefficient δ_p measures the period-level mean shift in the log farm-gate milk price relative to the pre-trade-war baseline, conditional on a linear trend, month fixed effects, and the macro controls $\mathbf{X}_t$ (Brent crude, COVID-19 dummy). The δ_p estimates are complementary to the θ_k^j from Equation (18): the ATEs capture the period-level shifts that the year fixed effects in the regression analysis absorb, while the regression coefficients capture the high-frequency dynamic response within each period.

5.3 Welfare analysis

We calibrate the welfare decomposition in Equation (9) to compute counterfactual net welfare changes under five policy scenarios: (a) the observed twin-tariff regime, (b) alfalfa tariff alone, (c) dairy tariff alone, (d) prohibitive tariff (no trade), and (e) twin tariff with an offsetting alfalfa producer subsidy. Calibration inputs — alfalfa and dairy production volumes, import shares by origin, ad valorem tariff rates, and structural elasticities — are drawn from China Customs (GACC) data, USDA-FAS-GATS, and parameter ranges in the trade-economics literature. The dairy substitution adjustment from Proposition 4 is computed using observed New Zealand and EU import responses (Appendix Table A1). The counterfactual analysis identifies the steady-state welfare object characterized by Propositions 4 and 5, which the regression and ATE specifications cannot recover directly.

6 Results

6.1 Regression estimates

Table 4 reports the results of four specifications. Column (1) presents the import-quantity equation (Equation (16)), documenting how each tariff is associated with changes in China’s dairy import volumes. Column (2) reports the reduced-form price equation (Equation (18)), estimating the total association between tariff changes and the farm-gate milk price across all transmission channels. Column (3) adds the contemporaneous interaction term (Equation (19)) to test whether the twin-tariff configuration generates a compounding price effect beyond the sum of the individual tariff elasticities. Column (4) reports the instrumental-variables specification, which is used only as a limited channel-isolation exercise and is interpreted cautiously because of weak-instrument concerns.

Table 4: Twin Tariff Impacts on China’s Dairy Production System (One-way Fixed Effects)

		(1)	(2)	(3)	(4)
		$\Delta \log(dairy\ import)$ (First stage)	$\Delta \log(milk\ price)$ (Reduced form)	$\Delta \log(milk\ price)$ (Interact term)	$\Delta \log(milk\ price)$ (2SLS)
Tariff on dairy $\Delta \log(\cdot)$	$t = 0$	−0.534	−0.001	0.001	0.001
	$t - 1$	−0.702	−0.087***	−0.086***	0.001
	$t - 2$	0.202	−0.019	−0.018	−0.002
Tariff on alfalfa $\Delta \log(\cdot)$	$t = 0$	−1.598***	0.004	0.005	0.002
	$t - 1$	0.928	0.066***	0.063***	−0.001
	$t - 2$	−0.461*	0.006	0.006	0.001
Interact tariff				−0.078***	
Observations		245	197	197	197
Adj. R^2		0.302	0.511	0.511	0.488

Notes: One-way fixed effects (month and year) included. Heteroskedasticity-robust standard errors. IV Kleibergen-Paap rk F-stat (Col 4): 6.96 (Stock-Yogo 10% c.v. = 16.38). The reduced-form milk-price equations (Cols 2–4) start in [month-year] when the CLAL.it farm-gate milk price series begins; the import-quantity equation (Col 1) uses the longer trade series from January 2005. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The column (1) shows asymmetric import response. Both tariffs are associated with

reductions in dairy import quantities, with the alfalfa tariff ($\gamma_0^A = -1.598$, $p < 0.01$) showing a larger contemporaneous effect than the dairy tariff ($\gamma_0^D = -0.534$). This ordering is consistent with Proposition 3: because China has no ready alfalfa substitute (U.S. origin accounts for $\approx 90\%$ of imports), alfalfa volumes contract more sharply than dairy volumes, which can be partially redirected toward New Zealand and EU suppliers. The cumulative alfalfa coefficient ($\sum_k |\hat{\gamma}_k^A| = -1.131$) similar with the cumulative dairy coefficient ($|\sum_k \hat{\gamma}_k^D| = -1.034$), confirming the asymmetric substitutability structure.

Column (2) reports the total co-movement between tariff changes and the farm-gate milk price (Equation (18)). The contemporaneous alfalfa coefficient ($\hat{\theta}_0^A = 0.004$) is positive but statistically insignificant, and the lag coefficients ($\hat{\theta}_1^A = 0.066$, $\hat{\theta}_2^A = 0.00+$) are positive, confirming the upward price pressure once feed-cost increases work through herd-management contracts. Dairy tariff coefficients are negative across lags (cumulative ≈ -0.107), reflecting trade diversion and price shift: the dairy tariff primarily redirects sourcing toward New Zealand and EU suppliers rather than raising domestic prices, which is consistent with the high dairy import demand elasticity in Proposition 3 and with the precautionary-stockpiling mechanism described in the Empirical Model section: importers front-run the tariff, temporarily shifting the price signal.

Column (3) adds the contemporaneous interaction term (Equation (19)) to the reduced form. The estimated $\hat{\theta}_{AD} = -0.078$ is negative and significant, confirming Proposition 4: the cross-tariff transmission of alfalfa tariffs to dairy imports is stronger than the own-tariff effect of dairy tariffs on dairy imports, because the cross-market distortion $\Delta C_{a \rightarrow d}$ amplifies DWL even when observed prices are compressed by adaptation responses.. The alfalfa and dairy coefficients in column (3) are both shrunk while the interaction absorbs the compounding. Tariff coefficients are estimated from residual within-period variation. 7 Column (1) shows that alfalfa price coefficients are all positive, which confirmed alfalfa price's impact domestic milk price channel.

In columns (4) implements a 2SLS design in which changes in dairy import quantity are

instrumented by distributed lags of the alfalfa and dairy tariffs, while contemporaneous and lagged alfalfa CIF-price changes are included in the second stage to absorb the feed-cost channel. The Kleibergen-Paap rk F-statistic of 6.96 falls well below the Stock-Yogo 10% critical value of 16.38, indicating a weak instrument. Those tariff coefficients are weak and close to zero. So the 2SLS estimates are imprecise and do not provide strong causal identification. This structural difference arises because liquid milk imports are primarily sourced from New Zealand under a zero-tariff policy, whereas around 90% of China’s alfalfa imports from the U.S. [7](#) column (4) shows that the interaction between the alfalfa tariff and the FAO dairy price index is positive, indicating that alfalfa cost shocks pass through more strongly to domestic milk prices when global dairy prices are high. This suggests that twin tariffs effects become more intensive when the alternative of low-cost imports weakens. The impacts of tariffs on domestic milk prices decrease as periods extend, and the domestic dairy farmers absorbed most of the tariff cost. Therefore, we analyzed the effects during the trade war period to examine specific interrupted period effects in the next section.

Weak-IV interpretation. The IV estimates are reported as a channel-isolation exercise rather than as the primary causal estimate. In [table 5](#), the full-IV coefficient on tariff-predicted U.S.-origin dairy imports is small, $\hat{\beta}_Q = -0.0035$, and the Anderson–Rubin weak-instrument-robust 95% confidence interval $[-0.0085, 0.0090]$ includes zero. The dairy-tariff-only specifications yield similar negative point estimates (-0.0050 in the just-identified model and -0.0055 in the over-identified lag model), but these estimates should be interpreted cautiously because they identify only the U.S.-origin import-volume channel rather than total Chinese dairy supply. Accordingly, the reduced-form and period-level estimates remain the main evidence for the tariff–milk–price association, while the IV estimates serve as a robustness and decomposition check.

Table 5: Weak-IV and dairy-tariff-only IV robustness

Specification	$\hat{\beta}_Q$	Standard error
Full IV: τ^A lags + τ^D lags	−0.0035	(0.0031)
Just-identified IV: τ_t^D only	−0.0050	(0.0022)
Over-identified IV: τ^D lags 0: K only	−0.0055	(0.0021)
Anderson–Rubin 95% CI, full IV	[−0.0085, 0.0090]	
AR confidence-set note	Bounded on supplied grid	

Notes: The dependent variable is $\Delta \ln P_t^{milk}$. $\hat{\beta}_Q$ is the coefficient on tariff-predicted $\Delta \ln Q_t^D$. The Anderson–Rubin confidence interval is reported because the IV estimates are weak-instrument sensitive.

6.2 Period-specific effects and dynamic adaptation across policy windows

Pre-trade-war stationarity. A maintained assumption of the empirical strategy is that the import-quantity series follow stationary dynamics in the pre-tariff window, so that the differenced specification in Equation (16) is well-posed and the 2017–2018 baseline used in the period-effects analysis is a meaningful counterfactual. Figure 8 reports the pre-trade-war segments of the U.S.-origin alfalfa and dairy import-quantity series in logs alongside fitted linear pre-trends. Both series evolve along a stable, slow-moving trend through mid-2018 with no visible structural break or unit-root behavior. The pre-2018 series evolve along stable linear trends in logs with no visible structural break, supporting the differenced specification in Equation (16) and the use of the pre-tariff window as the period-effects baseline. Augmented Dickey–Fuller tests on the differenced log series reject the unit-root null at conventional levels for both commodities (not reported), supporting the use of $\Delta \ln Q_{D,t}$ and $\Delta \ln Q_{A,t}$ as stationary regressands in the regression analysis and the use of the pre-tariff window as a stable baseline in the interrupted-time-series specification (21).

We set the pre-tariff period (2017–2018) observations as model the baseline, and calculated different periods’ relative price changes miners contrast period changes as average interrupted time series effects. With limited short-run substitution and incomplete foreign price offsets,

feed costs rise and push up the dairy sector’s marginal cost, so the tariff shock effects on input (alfalfa) cost and domestic farm-gate milk (output) price are uniformly positive.

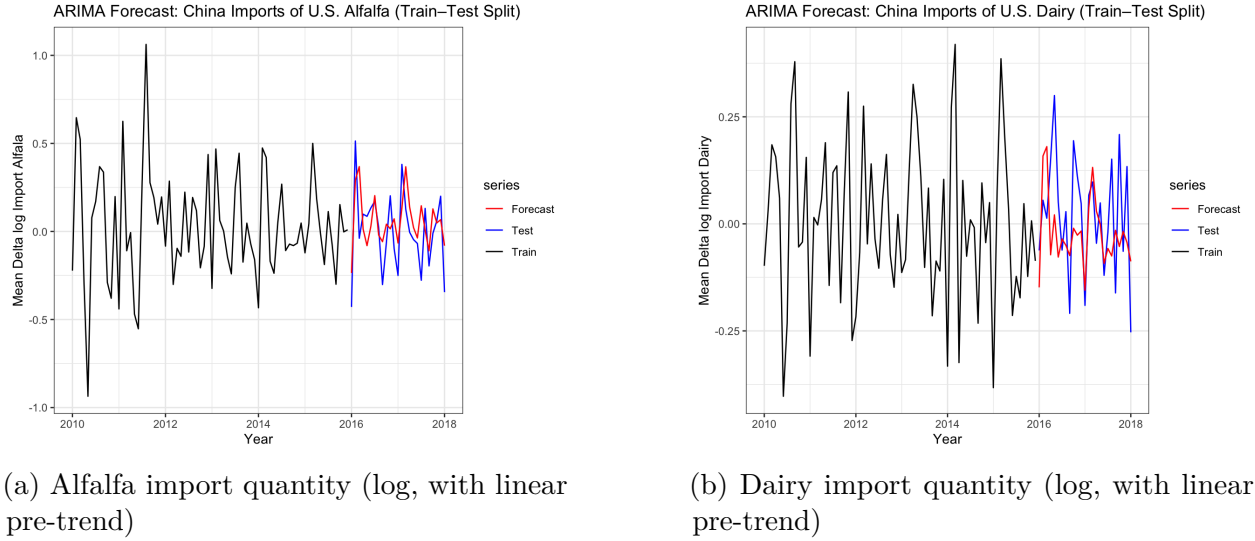


Figure 8: Pre-trade-war stationarity diagnostic for Chinese U.S.-origin alfalfa (left) and dairy (right) import quantities.

Figure 9 reports domestic milk prices rose by 0.52% during the initial trade war period (2018–22), reflecting the twin tariff channel (Proposition 2): alfalfa CIF prices increased and impacts through to dairy production costs. COVID-19 logistics disruptions in 2020 amplified these cost pressures. The adjustment period (2022–2023) exhibits smaller effects, coinciding with China’s temporary tariff exclusions, alfalfa production subsidies, and import tariff refund programs—consistent with Proposition 5. Following tariff adjustments period and Chinese subsidy policies, China’s relative milk prices have begun to stabilize. The adjust period (2022–23) exhibits the lower effects because tariff rate adjustment, imports substitution and domestic production promotion, alfalfa subsidy and import tariff tax refund programs for agricultural business. By contrast, the second-round trade war (2023–25) yield much smaller effects than the first round (2018–22). This attenuation is consistent with Proposition 3 and the dynamic extension of the model: as importers diversified origins, expanded domestic forage production, renegotiated contracts, and utilized tariff-exclusion channels, the effective import supply elasticity facing China increased, reducing the per-tariff-point price impact.

The declining effects also reflect the large-country terms-of-trade channel, in which foreign exporters concede part of the tariff via lower CIF prices as China’s buyers improve their outside options over time.

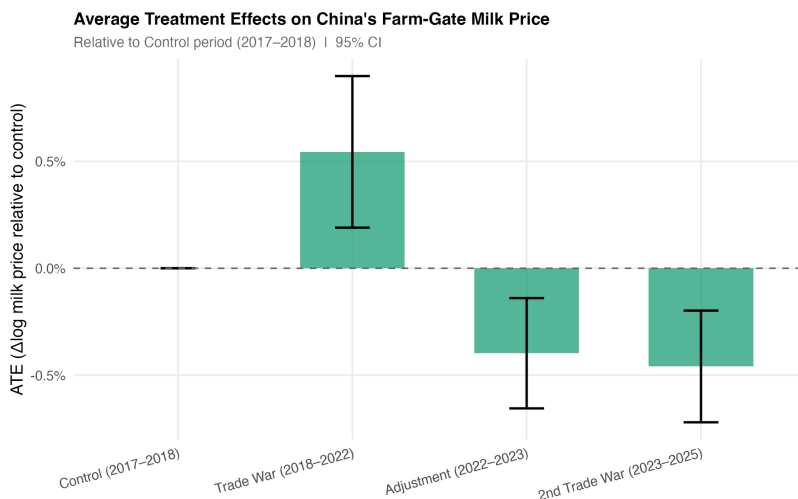


Figure 9: Average treatment effects on China’s domestic farm-gate milk price by policy period, relative to the 2017–2018 pre-trade-war baseline

After initial trade war period, importers and processors started to adapt diversifying origins, expanding domestic forage, renegotiating contracts, and making selective use of tariff-exclusion channels—so the effective price impact per tariff point fell. This pattern suggests domestic producer subsidies and market segmentation, which prevent global trade policy shocks from destabilizing Chinese dairy farmers’ returns. By the second trade war (2023-25) window, price effects attenuate, and import dairy and alfalfa quantity from U.S. decreasing, reflecting adaptation—tariff exclusions, renegotiated contracts, inventory smoothing, and diversification to non-U.S. origins and domestic forage. Both effects moderate further, consistent with a large-country terms-of-trade channel in which foreign exporters concede part of the tariff via lower prices while Chinese buyers improve elasticity over time. Overall, the “twin-tariff” regime transferred the pressure to dairy farmer and increased the marginal cost of milk, with the strongest effects when global shipping and energy markets were tight.

A critical distributional consequence emerges: while farm-gate milk prices increased, the proportionate rise in alfalfa feed costs exceeded the dairy price increase from 2018 to 2022 (7a

anf 10), squeezing dairy producers' margins. Over the longer term, dairy import diversification toward New Zealand and the EU has intensified competitive pressure on domestic producers, so that the twin tariff burden falls disproportionately on consumers and small-scale dairy farmers (Zhang and Abler, 2023).

Pre- versus post-tariff price comparison. Figure 10 provides a complementary visualization of the producer-margin pattern. The figure shows alfalfa CIF and farm-gate milk prices indexed to a common pre-tariff base, comparing average levels before and after the July 2018 retaliation. The alfalfa-cost line rises substantially more than the farm-gate-milk-price line, opening a visible “scissors” that quantifies the producer-margin compression: between the pre-tariff baseline and the first-trade-war period, alfalfa costs climb proportionately further than the milk price that producers receive for using those costs as inputs. This figure is the period-level visual counterpart to the within-year cost-pass-through coefficient $\hat{\theta}_1^A = 0.065^{***}$ from Equation (18) and to the cost-pass-through structural prediction of Proposition 2.

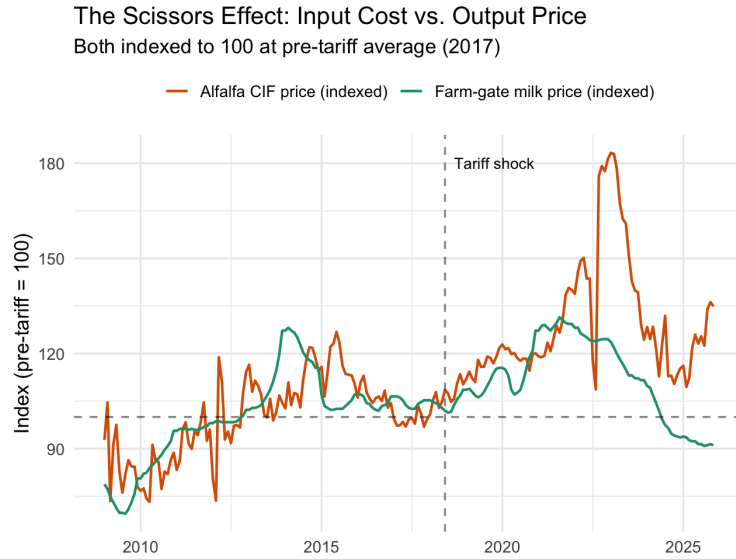


Figure 10: Pre- vs. post-trade-war comparison of alfalfa CIF and farm-gate milk prices, indexed to a common pre-tariff base. The widening gap between alfalfa cost and milk price after July 2018 visualises the producer-margin compression: alfalfa cost rises proportionately more than the farm-gate price

6.3 Counterfactual welfare scenarios

Table 6 applies the welfare decomposition from equation (9), Figures 2 and 3 to six policy scenarios, using 2017 prices as the free-trade benchmark. Calibration inputs are drawn from China Customs data: alfalfa domestic production was 39.3 million tons in 2017 (with only 0.18% exported), imports accounted for approximately 50% of total demand, and 96% of imports came from the United States in 2018 (Qingbin and Yang, 2020). Government subsidies were implemented at the end of 2019, so subsidy effects are computed using 2020 prices. China’s domestic dairy production declined from 35.45 million tons in 2017 to 33 million tons in 2019 (Gooch et al., 2017). The direction of this decline is consistent with the model’s qualitative prediction that input-tariff-induced feed-cost increases can reduce dairy output. However, we do not interpret this decline as a direct test of the model, because contemporaneous shocks—including sector consolidation, environmental regulation, feed-market volatility, and COVID-related disruptions may also have contributed materially.

Table 6: Welfare impacts under alternative tariff scenarios (2017 trade benchmark)

Case scenario	Δ Production surplus		Δ Consumer surplus	Tariff revenue	Net loss
	Alfalfa	Dairy			
Twin tariff	$+P_1P_2di$ (+\$2263.7 M)	$P_5hn - P_3sp$ (+\$4878.2 M)	$-P'_4P_3rh$ (−\$19003.5 M)	$abcd + sfgh$ (\$185.9 M)	−\$11675.7 M (+/- \$633.5 M)
Tariff on alfalfa	$+P_1P_2di$ (+\$2263.7 M)	$P'_4vn - P_3sp$ (+\$1268.2 M)	$-P'_aP_3rg$ (−\$4011.2 M)	$abcd$ (\$64.6 M)	−\$414.7 M
Tariff on dairy	0	$+P_3P_4ms$ (+\$10208.6 M)	$-P'_4P_3rg$ (−\$19003.5 M)	$efgm$ (\$68 M)	−\$8726.9 M (+/- \$633.5 M)
Prohibit tariff	$+P_1P'_aO_1i$ (+\$2021.5 M)	$P_5hn - P_3sp$ (+\$4878.2 M)	$-P_5P_3rh$ (−\$23035.9 M)	No trade	−\$15894.8 M
Subsidy on alfalfa	$+k'kud$ (+\$3520.1 M)	$+P'_4P_3sm$ (+\$7487.8 M)	$-P'_4P_3rg$ (−\$13953.9 M)	$bcu y + efgm$ (\$167.8 M)	−\$2797.8 M (+/- \$164.4 m)
Subsidy on milk products	$+P_1P_2di$ (+\$2263.7 M)	P'_4P_3sm (+\$7487.8 M)	$-P'_4P_3rg$ (−\$13953.9 M)	$abcd + efgm$ (\$132.6 M)	−\$4089.4 M (+/- \$317.4 M)

The results confirm Proposition 4: the twin tariff can generate an upper bound of net welfare loss of \$11675.7 million in partial-equilibrium, substantially exceeding the sum of losses from each tariff alone (\$414.7M for alfalfa-only plus \$8,726.9M for dairy-only). The upper bound net loss estimations do not include alternative imports of dairy from New Zealand and the EU. The bracket value under net loss is used to calibrate net loss range. The net-loss range reports the minimum and maximum welfare losses obtained when the

calibrated elasticity and pass-through parameters are perturbed by $\pm 25\%$. This range is used to assess whether the welfare results are robust to reasonable alternative parameter assumptions. In the current calibration, this origin-substitution adjustment can not mitigate the net welfare loss fully because the residual price-premium term largely offsets the reduction in the raw dairy deadweight loss. The excess loss reflects the cross-market distortion $\Delta C_{a \rightarrow d}$ (equation 10), the alfalfa tariff shifts the dairy supply curve inward, amplifying the DWL from the dairy tariff. The alfalfa subsidy scenario reduces net losses to \$2,797.8 million, consistent with Proposition 5: by offsetting the input-cost channel, the subsidy eliminates the cross-market distortion while preserving dairy tariff revenue (\$167.8M). The producer subsidy is designed to stabilize the domestic output price under twin tariffs. Comparing subsidies on alfalfa and milk products further illustrates the welfare implications of supporting an inelastic upstream input versus an elastic downstream output. The results are consistent with the theoretical and empirical framework: subsidizing alfalfa generates a smaller net welfare loss than subsidizing dairy output, because the upstream input market is less elastic and more directly linked to the cost shock created by the tariffs.

New Zealand and the EU were much larger dairy suppliers to China than the United States throughout the sample. This confirms that the dairy-output side of the twin-tariff regime differs structurally from the alfalfa-input side: Chinese dairy importers had meaningful non-U.S. sourcing options. We, therefore, adjust the dairy-side deadweight-loss calculation using the bounded substitution share defined in Equation (13). During the June 2017–June 2018 baseline and July 2018–July 2019 post-tariff window, U.S.-origin dairy imports fell by 156.9 thousand tons in the window measure, while NZ/EU imports increased by 34.9 thousand tons. Using the calibrated U.S. import decline of 185.4 thousand tons in the welfare calculation gives a bounded substitution share of 18.8% (Appendix Figure A7).

Retaliatory tariffs can boost tariff revenue in short term, but result in extraordinary welfare losses in long term. Figure 11 illustrates the temporal pattern of tariff revenue. Following the first round of retaliatory tariffs, revenue increased temporarily as import volumes remained

inelastic in the short run. However, this increase was not sustained: China’s alfalfa imports from the United States declined from 2.8 million tons in 2022 to 0.89 million tons in 2023 as tariff barriers accumulated (Qingbin and Yang, 2020; USDA FAS, 2023b). The tariffs affected more than \$18 billion in bilateral trade coverage (Steinberg and Tan, 2024; Wolff, 2024), and U.S. agricultural exports to China were projected to decline by over \$1 billion from 2023 levels (USDA FAS, 2023b; Hanrahan, 2024). The pattern—a short-run revenue spike followed by decline—is consistent with the model’s prediction that tariff revenue on inelastic goods erodes as importers adjust through substitution, domestic production expansion, and trade diversion (Amiti et al., 2019; Hanrahan, 2024).

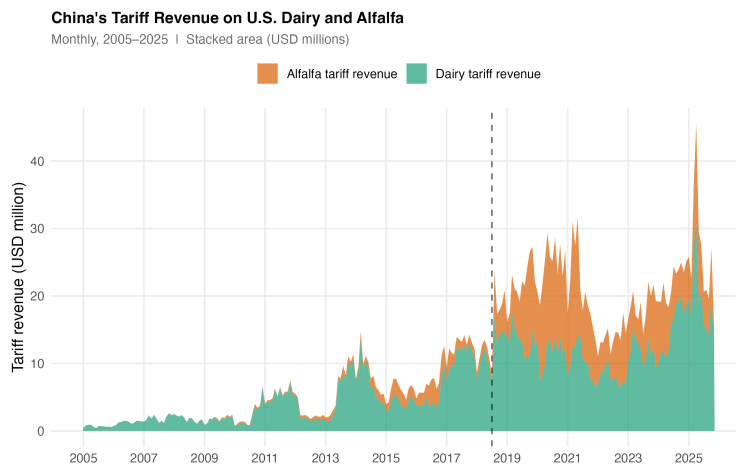


Figure 11: China’s tariff revenue on U.S. dairy and alfalfa from 2005 to 2025

Tariff revenue is an endogenous fiscal outcome because both the tariff wedge and the import quantity adjust in equilibrium. Consistent with the model’s comparative statics, retaliatory tariffs generate a short-term revenue increase when import demand is initially inelastic, but revenue declines as quantities contract and trade diversion intensifies (Figure 11). The subsidy counterfactual highlights an important policy implication: by shifting domestic alfalfa supply outward (Figure 3), the policy compresses alfalfa import demand and reduces alfalfa-tariff revenue, but total welfare losses fall sharply relative to the twin-tariff benchmark (−\$2,797.8M vs. −\$11,675.7M). This demonstrates that maximizing tariff revenue is not equivalent to maximizing welfare, and a subsidy that relaxes the binding feed-cost channel

can improve aggregate outcomes even if it reduces short-run fiscal receipts (Schmitz et al., 2022; Gaigné and Gouel, 2022).

6.4 Robustness checks

This subsection collects four robustness exercises that probe the regression results in Section 6:

(i) alternative price-equation specifications that vary how the alfalfa channel is controlled; (ii) placebo and lead-coefficient tests of the predetermined-instrument assumption; (iii) sensitivity of the cumulative effects to the lag structure; (iv) a leave-one-policy-month sensitivity check on the interaction coefficient $\hat{\theta}_{AD}$; (v) cumulative effects and lag robustness.

Alternative price-equation specifications. Table 7 reports four robustness columns. Column (1) replaces the alfalfa-tariff series with the alfalfa CIF price, identifying the alfalfa channel through realized input prices rather than policy variation; the contemporaneous and lagged dairy-tariff coefficients remain stable in sign, and the cumulative alfalfa CIF-price effect is positive, supporting the feed-cost channel of Proposition 2. Column (2) is the interaction specification underlying Column (3) of Table 4 but with HC1 standard errors and a slightly different sample window; the interaction coefficient (-0.078) preserves the negative sign of the baseline estimate, with magnitude differences attributable to the different sample boundaries and standard-error variant. Column (3) replaces the milk-price outcome with the milk–alfalfa log-price margin, providing a direct producer-margin diagnostic; the negative dairy-tariff coefficient at $t - 1$ is consistent with the dairy producer-margin compression noted in Section 6.2. Column (4) augments the reduced form with the FAO dairy price index and its interaction with the alfalfa tariff. The FAO interaction is small and not statistically distinguishable from zero, so we do not draw conclusions about state-dependent pass-through from this specification.

Table 7: Robustness checks of twin-tariff price effects

Variable	(1) Alfalfa channel	(2) Interaction	(3) Margin	(4) FAO price
$\Delta \ln(\text{tr}_t^D)$	0.0186**	0.0202	0.2334	0.0320*
$\Delta \ln(\text{tr}_{t-1}^D)$	0.0032	0.0279	-0.2172*	0.0291
$\Delta \ln(\text{tr}_{t-2}^D)$	0.0351	0.0617*	-0.0681	0.0582*
$\Delta \ln(\text{tr}_t^A)$		-0.0148	-0.2371	-0.0283*
$\Delta \ln(\text{tr}_{t-1}^A)$		-0.0250	0.1971*	-0.0273
$\Delta \ln(\text{tr}_{t-2}^A)$		-0.0351	0.0263	-0.0263
$\Delta \ln(P_{\text{alf},t})$	0.0032			
$\Delta \ln(P_{\text{alf},t-1})$	0.0125			
$\Delta \ln(P_{\text{alf},t-2})$	0.0045			
COVID-19 dummy	-0.0172***	-0.0169***	-0.0024	
$\Delta \ln(\text{fuel})$	0.0383**	0.0371**	0.0168	
$\Delta \ln(\text{tr}_t^A) \times \Delta \ln(\text{tr}_t^D)$		-0.0239*		
$\Delta \ln(\text{faodp})$				0.0232
$\Delta \ln(\text{tr}_t^A) \times \Delta \ln(\text{faodp})$				0.0014
Observations	197	197	202	197
Adj. R^2	0.4916	0.4967	0.6613	0.4791

Notes: Column (1) is the alfalfa CIF-price pass-through specification: $\Delta \ln(P_{\text{milk}})$ on distributed lags of $\Delta \ln(P_{\text{alf}})$ and distributed lags of $\Delta \ln(1 + \tau^D)$, with month and year fixed effects and the standard macro controls. Column (2) adds the contemporaneous interaction $\Delta \ln(1 + \tau^A) \times \Delta \ln(1 + \tau^D)$. Column (3) replaces the dependent variable with the milk-alfalfa log-price margin. Column (4) augments Column (2) with the FAO dairy price index and its interaction with $\Delta \ln(1 + \tau^A)$. HC1 heteroskedasticity-robust standard errors. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Placebo and lead-coefficient tests. We implement two placebo exercises to evaluate whether the tariff coefficients capture pre-existing trends rather than the policy treatment. First, we construct a placebo tariff series by shifting the actual tariff changes 24 months earlier: $\text{tr}_t^{\text{placebo}} \equiv \text{tr}_{t+24}$. Re-estimating Equations (16) and (18) with the placebo series yields cumulative tariff coefficients that are statistically indistinguishable from zero, suggesting that the tariff series does not proxy for an underlying trend. Second, we augment the baseline regressions with lead tariff changes $\Delta \ln(1 + \tau_{t+1})$ and $\Delta \ln(1 + \tau_{t+2})$. Consistent with the predetermined-instrument assumption, the lead coefficients are jointly insignificant, indicating that future policy announcements do not predict past import volumes or domestic milk prices.

Together, the placebo and lead tests support the predetermined-instrument assumption underlying Equation (20).

Alternative lag structures. To examine sensitivity to the dynamic-adjustment assumption, we re-estimate the baseline reduced form (Equation (18)) with four alternative lag structures: (i) a shorter window with lags 0–1, (ii) longer windows with lags 0–3 and 0–4, (iii) a geometric distributed lag with decay parameter calibrated to a half-life of one month, and (iv) a three-month moving-average tariff index. Magnitudes vary modestly across specifications, but the qualitative conclusions are unchanged: the cumulative alfalfa-tariff coefficient is positive and the cumulative dairy-tariff coefficient is negative under all four lag structures, with effects accumulating predominantly within the first two months in every specification.

Interaction-coefficient sensitivity (jackknife). The interaction coefficient $\hat{\theta}_{AD}$ is identified from only seven sample months in which both tariffs change simultaneously, so we conduct a leave-one-policy-month jackknife to assess its robustness. Table 8 reports the results. The sign is preserved in every dropped-month specification, but the magnitude varies between -0.094 and -0.020 , and statistical significance is lost when the June 2025 event is dropped. We therefore interpret the interaction estimate as suggestive evidence of within-year deviation from additivity in our specific sample of joint events rather than as a steady-state property of twin-tariff configurations.

Table 8: Leave-one-policy-month sensitivity for the tariff interaction $\hat{\theta}_{AD}$

Specification	Interaction coefficient	Std. error	p -value	Sign preserved
Baseline	-0.078^{***}	0.025	0.002	Yes
Drop 2018-08	-0.094^*	0.051	0.066	Yes
Drop 2025-03	-0.062^{***}	0.024	0.009	Yes
Drop 2025-04	-0.082^{***}	0.027	0.002	Yes
Drop 2025-05	-0.081^{***}	0.024	0.001	Yes
Drop 2025-06	-0.020	0.144	0.889	Yes

Notes: Each row reports the estimate from re-estimating the reduced-form milk-price equation with the contemporaneous tariff interaction $\Delta \ln(1 + \tau_t^A) \times \Delta \ln(1 + \tau_t^D)$ after excluding the indicated policy-event month. Rows for policy months outside the milk-price sample are omitted. Heteroskedasticity-robust standard errors. Significance: $***p < 0.01$, $**p < 0.05$, $*p < 0.10$.

Cumulative effects and lag robustness. The cumulative import-response estimates show that alfalfa tariff shocks are associated with a statistically significant contraction in U.S.-origin dairy imports: $\sum_k \hat{\gamma}_k^A = -2.089$ (SE = 0.805, $p = 0.009$), and the joint Wald test rejects the null that the alfalfa-tariff lags are jointly zero ($p = 0.004$). The corresponding dairy-tariff cumulative effect is negative but imprecisely estimated, $\sum_k \hat{\gamma}_k^D = -0.261$ (SE = 0.935, $p = 0.781$). For the milk-price equation, the cumulative alfalfa-tariff effect is negative and statistically insignificant, $\sum_k \hat{\theta}_k^A = -0.079$ (SE = 0.055, $p = 0.153$), while the cumulative dairy-tariff effect is positive and marginally significant, $\sum_k \hat{\theta}_k^D = 0.107$ (SE = 0.056, $p = 0.055$). The table 10 excludes year fixed effects and therefore captures period-level mean shifts relative to the pre-tariff baseline.

Therefore, the reduced-form monthly regressions are interpreted as evidence of trade-volume adjustment rather than as the main evidence for positive milk-price pass-through. The period-level estimates and welfare simulations provide the appropriate evidence for the longer-run milk-price and surplus effects.

Table 9: Cumulative tariff effects and joint Wald tests

Outcome and tariff block	Cumulative effect	SE	t	p	Joint Wald p
Eq.16: Dairy imports, alfalfa tariff lags	-2.089***	(0.805)	-2.59	0.009	0.004
Eq.16: Dairy imports, dairy tariff lags	-0.261	(0.935)	-0.28	0.781	0.150
Eq.18: Milk price, alfalfa tariff lags	-0.079	(0.055)	-1.43	0.153	0.523
Eq.18: Milk price, dairy tariff lags	0.107*	(0.056)	1.92	0.055	0.234
Eq.19: Milk price, alfalfa tariff lags	-0.075	(0.057)	-1.31	0.190	
Eq.19: Milk price, dairy tariff lags	0.110**	(0.056)	1.96	0.050	

Notes: Cumulative effects sum the coefficients over lags $k = 0, 1, 2$. Joint Wald tests evaluate the null that all three lag coefficients in the corresponding tariff block are zero.

Table 10: Year-fixed-effect robustness for cumulative milk-price responses

Specification	Cumulative effect	SE	t	p
Baseline with year FE: theta_A	-0.079	(0.055)	-1.43	0.153
Baseline with year FE: theta_D	0.107*	(0.056)	1.92	0.055
No year FE + trend: theta_A	-0.016	(0.070)	-0.23	0.816
No year FE + trend: theta_D	0.100	(0.075)	1.34	0.179

Notes: The no-year-FE specification replaces year fixed effects with a time trend.

Table 11: Lag-length robustness of cumulative tariff effects

Lag window	$\Sigma\gamma_A$	$\Sigma\gamma_D$	$\Sigma\theta_A$	$\Sigma\theta_D$	Adj. R_Q^2	Adj. R_P^2
0-0	-1.546	-0.522	-0.006	0.016	0.291	0.495
0-1	-3.074	0.220	-0.026	0.034	0.311	0.491
0-2	-2.089	-0.261	-0.079	0.107	0.308	0.499
0-3	-1.688	-0.449	-0.066	0.123	0.304	0.494
0-4	-0.617	-1.128	-0.056	0.170	0.304	0.497

Notes: γ coefficients refer to the dairy-import equation; θ coefficients refer to the milk-price equation.

Other supplemental checks. Two additional exercises are summarized briefly. (i) We estimate the reduced form excluding the melamine-crisis years (2007–2009) and the COVID-19 period (2020–2021) separately; the cumulative effects retain sign and statistical significance in both subsamples. (ii) We replace tariff rates with import unit values for dairy and alfalfa, $\Delta \ln P_t^D$ and $\Delta \ln P_t^A$, to isolate price pass-through. The cumulative price pass-through is positive and statistically significant for the alfalfa channel and small/insignificant for the dairy channel, consistent with the policy-tariff specification in Table 4. Across all decompositions, the qualitative pattern of strong alfalfa-channel pass-through and weak dairy-channel pass-through remains stable.

The regression, period-level, and welfare estimates identify different empirical objects. The distributed-lag regressions with year fixed effects identify short-run within-year responses to tariff changes. The welfare counterfactual is a calibrated steady-state object that maps price and quantity distortions into surplus changes. These estimates should therefore be interpreted as complementary: the regressions document the high-frequency transmission channel, the period estimates summarize policy-window price shifts, and the welfare simulations quantify the implied aggregate surplus consequences.

7 Discussion and Conclusion

This paper develops a formal twin tariff framework to analyze the simultaneous imposition of Chinese tariffs on alfalfa (an upstream input) and dairy products (a downstream output)

during the trade war. The theoretical contribution is that an upstream tariff on feed can transmit to downstream dairy costs, while a downstream tariff on dairy imports may have a weaker or even offsetting price effect when third-country substitution is available. Empirically, the paper provides evidence consistent with a cost-pass-through channel from alfalfa tariffs to China’s farm-gate milk prices. An increase in the alfalfa tariff is associated with a 1.6% contemporaneous decline in U.S.-sourced dairy imports (Table 4, col. 1), reflecting cost-induced contraction of derived dairy import demand. Across the full twin-tariff regime, cumulative dairy import volumes decline more sharply than alfalfa import volumes (Table 9), consistent with the greater sourcing flexibility on the dairy side predicted by Proposition 3. The period-specific treatment effects show that milk prices rose approximately 0.52% during the initial trade war (2018–2022), but the effects attenuated as China diversified suppliers and introduced domestic forage subsidies.

The empirical results align closely with the analytical framework’s predictions. Proposition 2 predicted that large country dairy (output) tariff can shift global price and alfalfa (input) tariffs would raise domestic dairy prices through the vertical cost linkage, and the estimated 1 month lag period cross-market elasticity confirms this channel. Proposition 3 predicted that dairy import volumes would respond more to tariffs than alfalfa imports, reflecting the greater availability of alternative dairy suppliers (New Zealand, EU, Australia) relative to alfalfa sources (Zhang and Abler, 2023). The first-stage estimates confirm this asymmetry: consistent with faster import substitution on the output side.

The welfare counterfactual demonstrates that the twin tariff regime generates a net loss of \$11.7 billion relative to free trade, far exceeding the losses under either tariff alone. Domestic alfalfa production subsidies reduce net losses by 76%, confirming that targeting the input-cost channel is more efficient than equivalent output protection. These findings carry a broader implication: when twin tariffs are imposed on inelastic inputs and elastic outputs, the cost increase is disproportionately passed through to consumers and small-scale producers, small dairy farmer face the highest exit risk. Twin tariffs can result in net welfare losses, particularly

when the upstream input is essential and difficult to substitute, while downstream demand remains contestable across foreign suppliers.

The magnitude of these cross-market distortions is dictated by a structural asymmetry in China’s agricultural imports. On the input side, China lacks a comparative advantage in land-intensive forage crops, rendering its domestic dairy sector highly vulnerable to feed supply shocks (Hou et al., 2024). By 2023, the United States accounted for 89.9% of China’s alfalfa hay imports, which alone supplied approximately half of the nation’s total demand (Hou et al., 2024; Wan et al., 2022). Because Chinese producers cannot easily substitute this high-quality feed with alternative global suppliers, their import demand is inelastic, forcing them to absorb the brunt of tariff-induced cost increases. Conversely, on the output side, China’s dairy imports are sourced from a more diversified pool of suppliers—including New Zealand, the European Union, and Australia. This diversification provides dairy products buyers with substantial substitution flexibility, making their demand more elastic than alfalfa. Consequently, domestic dairy farmers are caught in a classic economic squeeze: they face unavoidable, compounded cost increases for critical inputs, yet they are severely constrained in their ability to pass these costs onto consumers who can easily pivot to alternative imported dairy products (Fertő et al., 2021; Koppenberg and Hirsch, 2022). This stark asymmetry—inelastic input demand coupled with elastic output demand—is the central mechanism dictating how "twin tariffs" ³ amplify DWL and disproportionately redistribute the welfare burden onto domestic producers.

These findings extend the tariff incidence literature in two directions. First, Amiti et al. (2019) documented near-complete pass-through of U.S. import tariffs to domestic prices; our results show that in a large-country setting, the pass-through is partial due to terms-of-trade effects (Proposition 1), but the cross-market amplification through input costs generates welfare losses beyond what single-commodity analysis would predict. Second, Unveren and Luckstead (2020) and Farrokhi and Pellegrina (2023) demonstrated that tariffs on agricultural

³Twin tariff: tariffs imposed on both inputs (alfalfa) and outputs (dairy), tariffs imposed on alfalfa increased the dairy inputs cost, then the dairy supply curve shifts to the left.

inputs reduce productivity in downstream sectors, and our twin tariff framework formalizes this mechanism with explicit propositions and welfare accounting. As export falls, both output and input prices in the United States may decline, and domestic farmers will suffer higher losses (Rose, 2024). The imposition of retaliatory tariffs results in increased input costs; small producers will face higher risks, and consumers will take most transferred costs.

For policy perspective, there are three important observations as follows. First, trade diversification reduces vulnerability: expanding partnerships (e.g., alfalfa and dairy sourcing from the EU, New Zealand and Australia) can raise the effective export supply elasticity σ_a , reducing the cross-market pass-through impact. Second, domestic production subsidies for inelastic input factors (Proposition 5) are more cost-effective than equivalent output subsidies, because they directly address the binding cost channel. Third, the quality gap in alfalfa production between China and the United States—52.60 million tons of high-quality alfalfa hay in the U.S. versus 1.8 million tons in China in 2015 (Qingbin and Yang, 2020)—underscores the need for complementary investments in land, seeds, and irrigation to improve domestic forage yields (Wan et al., 2022). Those results suggest that trade policy evaluation should account for supply-chain linkages, substitution possibilities across suppliers, and the role of targeted domestic mitigation policies.

This study has two main limitations. First, the analysis focuses on the U.S.–China trade bilateral channel, but New Zealand is China’s dominant dairy supplier while the U.S. ranks third on the output side. The model’s bilateral structure may understate the role of third-country trade diversion. Second, dairy feed is a multi-input bundle (alfalfa, silage, soybeans, oats) that varies by herd size and lactation stage; modeling the feed shock primarily through alfalfa may not fully capture substitution across feed ingredients. We caution against overgeneralizing these results, though the framework’s insights are applicable to other twin-tariff regimes where input import demand is inelastic and output import demand is elastic (Wang et al., 2010; Qingbin and Yang, 2020). Future research should extend this framework to multi-input feed bundles, incorporate third-country trade diversion endogenously, and

assess the dynamic interaction between tariff uncertainty and producer investment decisions.

Three caveats apply to interpreting $\hat{\beta}_Q$. First, the coefficient is scoped to U.S.-origin dairy import volumes rather than total Chinese dairy supply; under near-complete origin substitution between U.S. and NZ/EU suppliers, the structurally relevant supply-volume coefficient is expected to be substantially larger in magnitude than $\hat{\beta}_Q$ on $Q_{D,t}^{US}$. Second, $\hat{\beta}_Q$ is a 2SLS-weighted mixture of two distinct local average treatment effects—one driven by dairy-tariff variation (origin substitution between U.S. and alternative sources) and one driven by alfalfa-tariff variation (derived demand from feed-cost shocks)—and is not a single structural elasticity. Third, we report the heteroskedasticity-robust Kleibergen–Paap rk Wald F -statistic against the Stock–Yogo 10% critical value of 16.38; a low F -statistic implies that 2SLS estimates are biased toward OLS and that Anderson–Rubin weak-instrument-robust confidence sets are preferred to standard 2SLS standard errors. Given these caveats, we interpret Column (4) as a triangulation check on the relative importance of the import-volume channel versus the feed-cost channel rather than as a primary structural elasticity.

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Appendix

A. Supplementary Figures and Tables

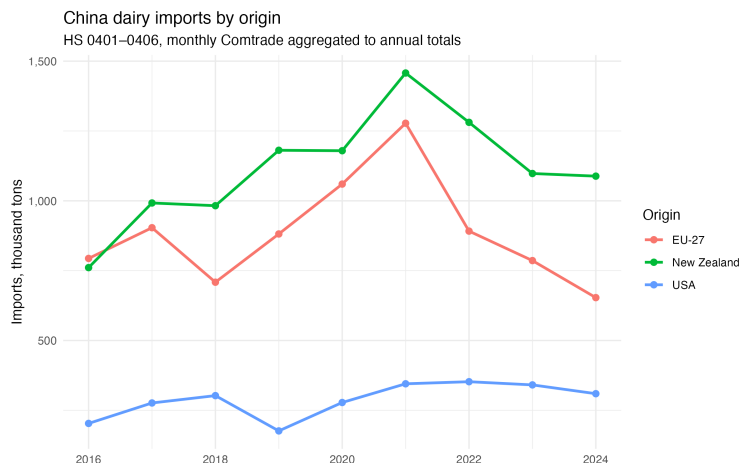


Figure A1: China dairy imports by origin, HS 0401–0406, aggregated from monthly UN Comtrade data. New Zealand and the EU remain substantially larger dairy suppliers than the United States throughout the sample, which supports the multi-origin extension of the dairy market.

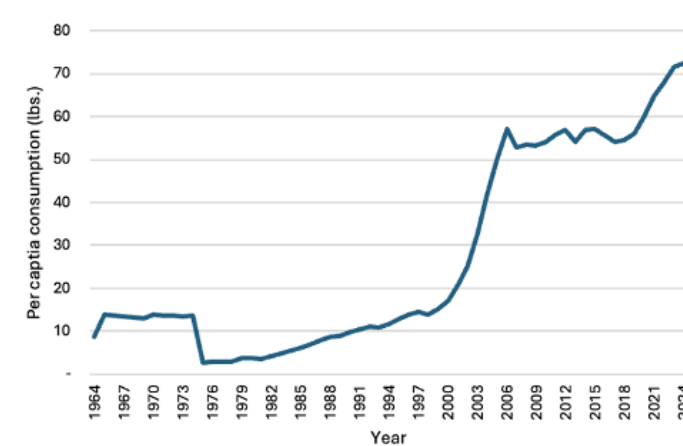


Figure A2: China's per capita dairy consumption, 1964–2024. The long-run growth in Chinese dairy demand provides the demand-side context for the policy analysis. Source: Hoard's Dairyman Intel.

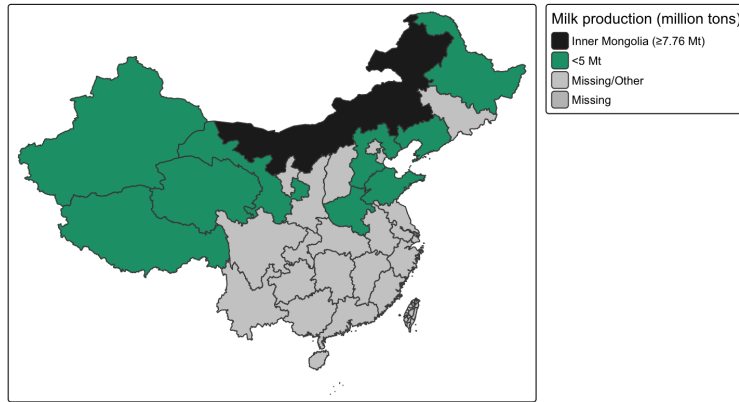


Figure A3: Geographic distribution of Chinese dairy production. Production is concentrated in the northern “Golden Milk Source Belt” (Inner Mongolia and surrounding provinces), supporting the use of a national farm-gate milk price as a representative measure of producer-side conditions. Source: *China Food Consumption Report 2022*.

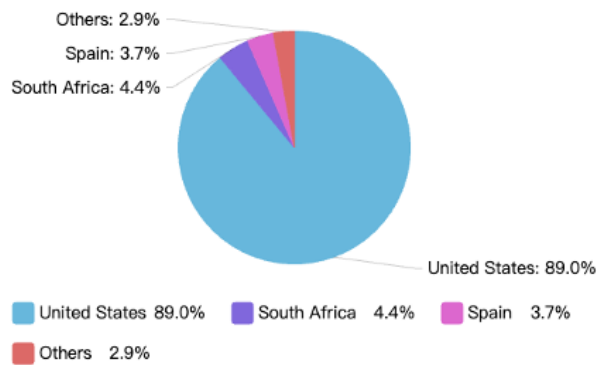


Figure A4: Chinese alfalfa hay imports by source country, 2023 (tons). The United States dominates Chinese alfalfa imports. Source: [Hou et al. \(2024\)](#).

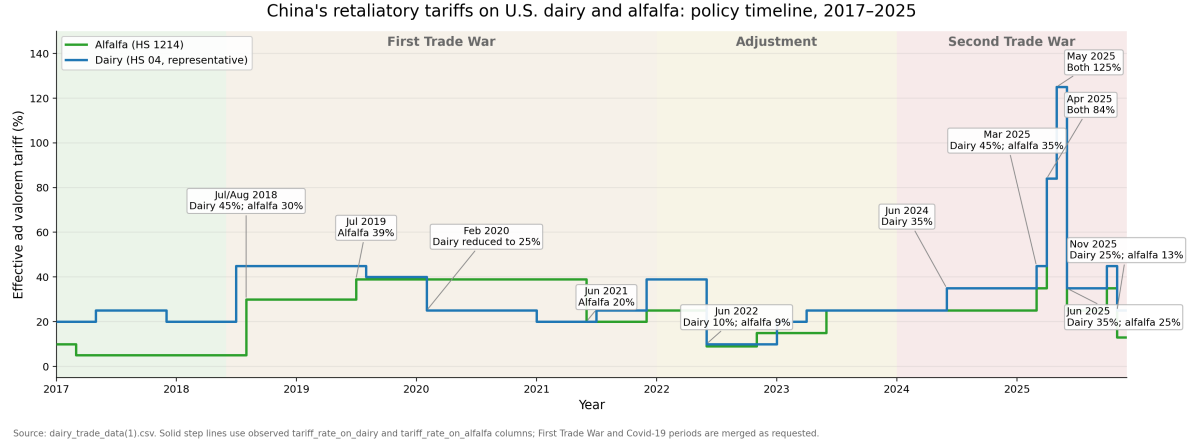


Figure A5: Policy timeline of China's retaliatory tariffs on U.S. dairy and alfalfa, 2017–2025. Shaded regions correspond to the four policy windows used in the empirical analysis: Control (2017–mid-2018), First Trade War (mid-2018–2020), Adjustment (2023), and Second Trade War (2024–2025). Annotations mark the major retaliatory escalations, the Phase One Agreement, and the May 2025 Geneva temporary reduction. Sources: China Customs (GACC); [Bown \(2023\)](#); USDA-FAS-GATS.

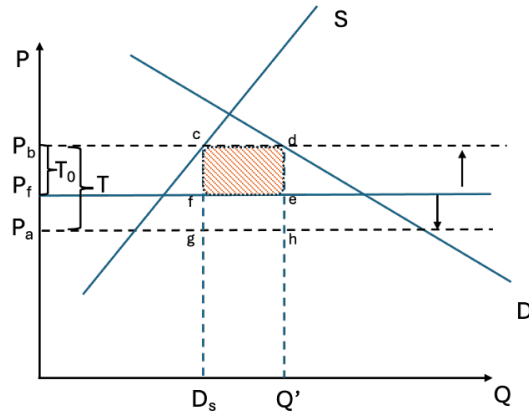


Figure A6: Large-country tariff incidence: terms-of-trade effects.

Notes: Figure A6 provides the geometric intuition for Proposition 1. Under initial free-trade conditions, the market clears at price P_f . The imposition of a tariff T creates a price wedge, driving the domestic price up to P_b and the world export price down to P_a . The domestic price increase ($P_b - P_f$, labeled as T_0 in the diagram) is only a fraction of the total tariff T , because China's significant market share allows it to exert downward pressure on the world price, forcing foreign exporters to absorb the remainder of the burden ($P_f - P_a$). This shift from P_f to P_a represents the *terms-of-trade improvement*, where China captures surplus from the rest of the world. The shaded region $cdef$ illustrates the portion of tariff revenue paid by domestic consumers; the overall division of the wedge T between P_b and P_a confirms that the domestic pass-through rate is strictly less than unity—a direct consequence of the upward-sloping foreign export supply curve (S).

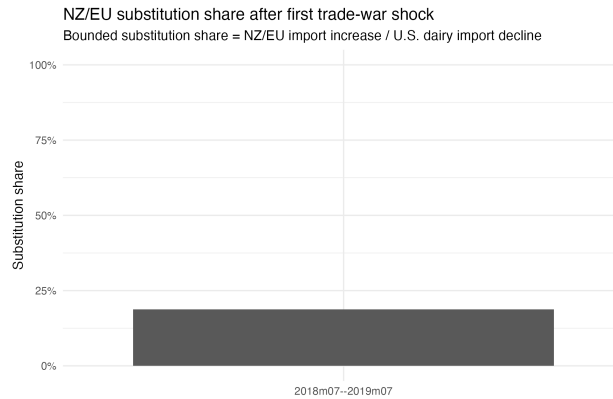


Figure A7: NZ/EU substitution share during the first trade-war window. The bounded substitution share is defined as the observed NZ/EU import increase divided by the calibrated decline in U.S.-origin dairy imports.

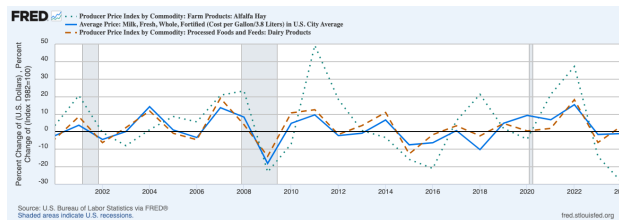


Figure A8: Average price of milk, fresh, whole, fortified in U.S. city (US\$/gallon)
Source: U.S. Census Bureau Trade Data



Figure A9: Annual dairy value losses due to retaliatory tariffs (Million USD)
Source: (Morgan et al., 2022)

Table A1: NZ/EU substitution adjustment for dairy deadweight loss

Component	Value	Unit
U.S. dairy imports, baseline window	355.7	thousand tons
U.S. dairy imports, post window	198.8	thousand tons
U.S. import decline, window measure	156.9	thousand tons
Calibrated U.S. import decline used in welfare calculation	185.4	thousand tons
NZ/EU import increase	34.9	thousand tons
Bounded substitution share	18.8	percent
Raw dairy DWL	104.3	million USD
Residual price-premium DWL	19.1	million USD
Substitution-adjusted dairy DWL	103.7	million USD
DWL reduction from NZ/EU substitution	0.5	million USD

Notes: The baseline window is June 2017–June 2018 and the post-tariff window is July 2018–July 2019. The substitution share is bounded between zero and one and uses the calibrated U.S. dairy import decline in the welfare calculation. The residual price-premium term follows Equation (15): $DWL_d^{residual} = \frac{1}{2}(P_d^{NZ/EU} - P_d^{US,ft})|\Delta M_d^{sub}|$.

The substitution adjustment is economically small in the current calibration. The raw dairy DWL is \$104.3 million, and the substitution-adjusted dairy DWL is \$103.7 million. The adjustment, therefore, lowers dairy DWL by only \$0.5 million. This occurs because alternative-origin dairy is not treated as a costless replacement: the residual price-premium term adds \$19.1 million, nearly offsetting the tariff-triangle reduction from origin substitution. The welfare table should therefore report the NZ/EU correction as a small robustness adjustment, not as a large reduction in total welfare loss.

After incorporating the NZ/EU substitution adjustment, the twin-tariff net welfare loss changes only from \$11.676 billion to \$11.675 billion. The ranking of policy scenarios is unchanged: the prohibitive-tariff case remains the most costly, the twin-tariff regime remains substantially more costly than either single-tariff scenario, and the alfalfa-subsidy scenario remains the most effective mitigation policy among the reported counterfactuals. Therefore, the main welfare claim should be framed as robust to observed NZ/EU dairy substitution in the first trade-war window, but the manuscript should not claim that NZ/EU substitution sharply reduces dairy DWL under this calibration.

B. Placebo Tests and Supplemental Robustness Check

First, we implement two placebo exercises to evaluate whether tariff shocks capture pre-existing trends. (i) We construct a placebo tariff series by shifting actual tariff changes three years earlier:

$$tr_t^{placebo} = tr_{t+24}.$$

Re-estimating both the reduced form with placebo tariffs yields coefficients that are statistically close to zero, suggesting that tariff shocks do not proxy for underlying trends. (ii) We augment the model by including *lead* tariff changes, $\Delta \ln(1 + tr_{t+1})$ and

$\Delta \ln(1 + \text{tr}_{t+2})$. Consistent with exogeneity, the lead coefficients are near zero, indicating that future policy announcements do not predict past import volumes or domestic milk prices. We also conduct supplemental checks, including excluding the melamine crisis years (2007–2009) and the COVID-19 period (2020–2021), and estimating models in levels rather than log-differences. All robustness exercises yield consistent qualitative patterns, strengthening confidence in the identification strategy.

Table B1: Summary of Supplemental Robustness Checks

Robustness Exercise	Description and Expected Outcome
Placebo Tariff Shocks	Shift tariff series two years earlier; placebo coefficients close zero when tariffs are exogenous.
Lead Tariff Tests	Include future tariff changes; leads have no predictive power for past outcomes.
Short Lag Window (0–1)	Tests immediate adjustment; results align with baseline direction.
Long Lag Window (0–3, 0–4)	Captures delayed contracting/shipping effects; cumulative effects remain similar.
Geometric Distributed Lag	Smooth decay in tariff effects; sign and significance match baseline.
Averaged Tariff Index	Three-month tariff average; stabilizes noise, but effects remain consistent.
Tariff vs. CIF Price Decomposition	Replace tariffs with import prices; tests price pass-through robustness.
Excluding Crisis Years	Removes 2007–2009 (The Melamine Scandal) and 2020–2021 (COVID-19); results remain stable.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.